

City of Detroit 2023 - 2025 Models

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Model I

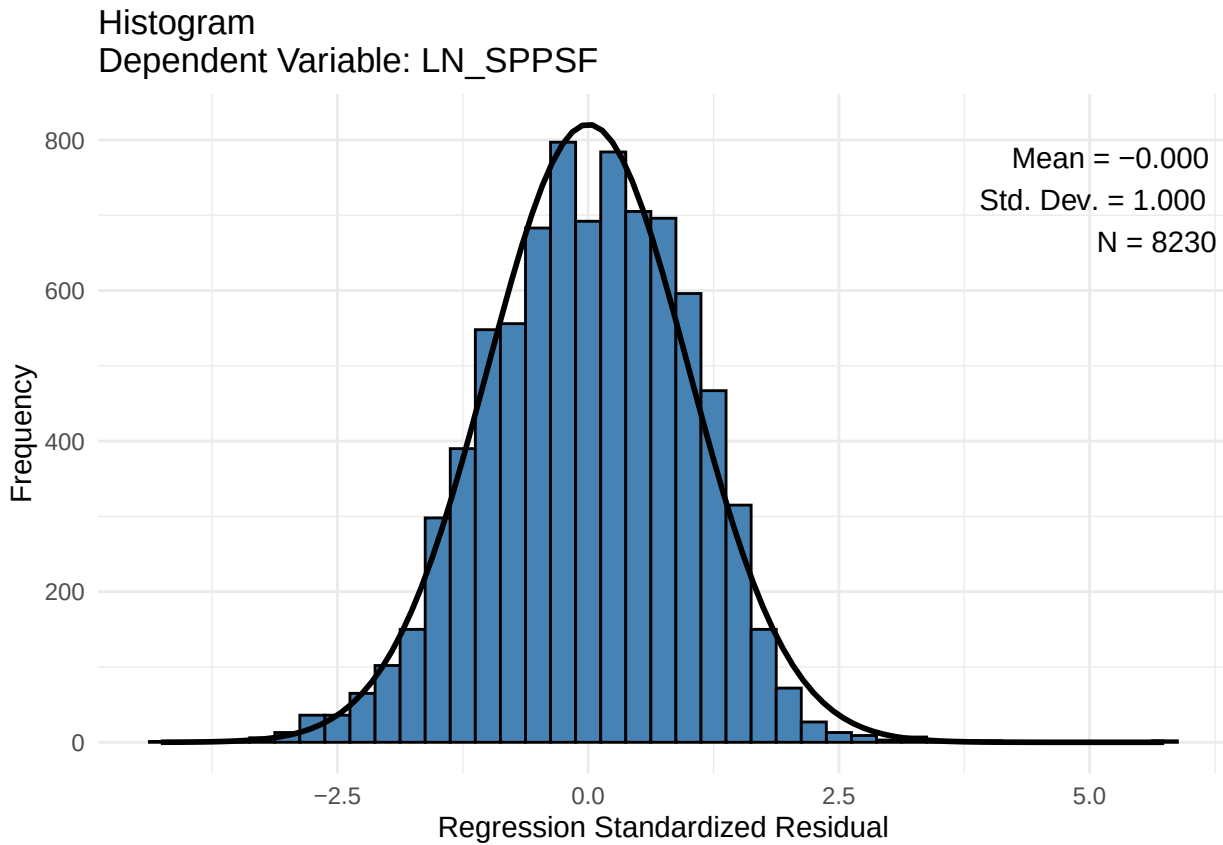


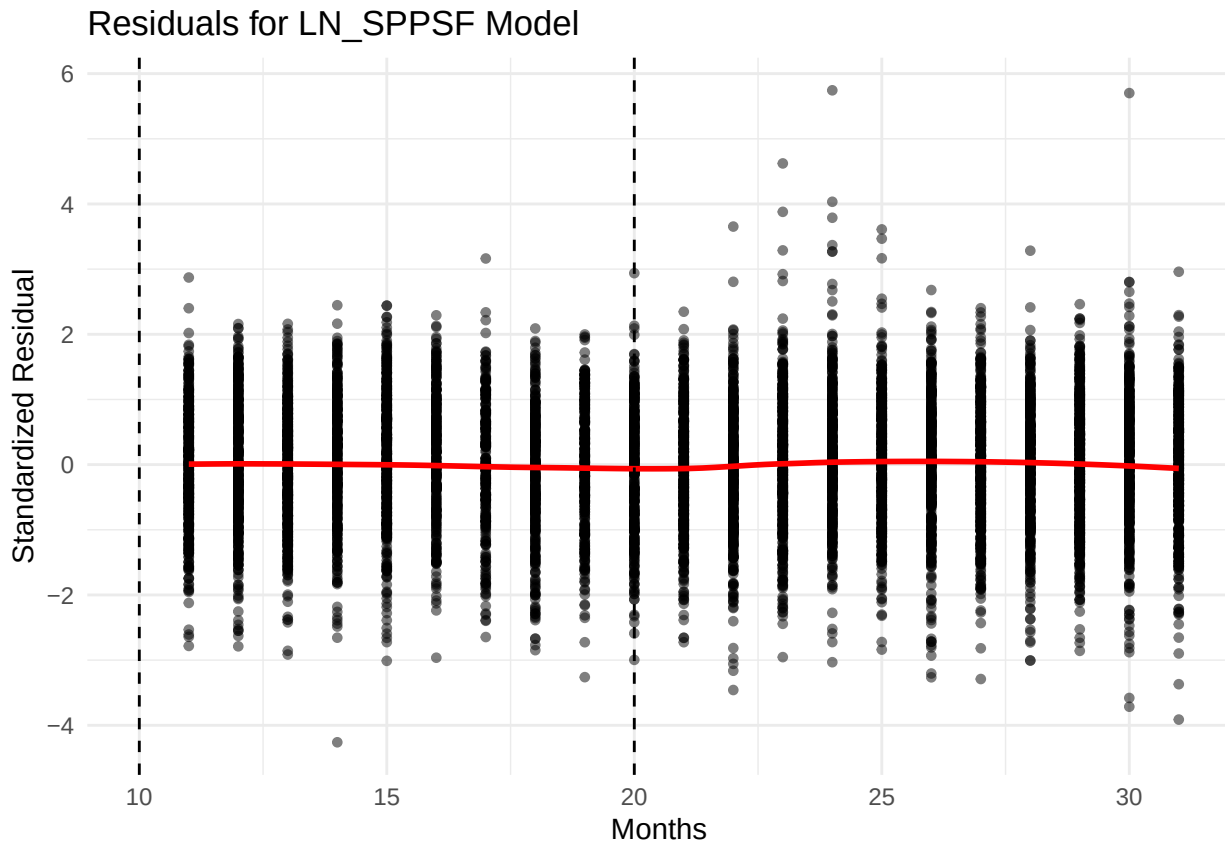
Table 1: LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
8230	8210	21	0.5083213	0.1302076	0.1281947	0.5077032	0.4103535	64.68601	0	12240.18	12387.51

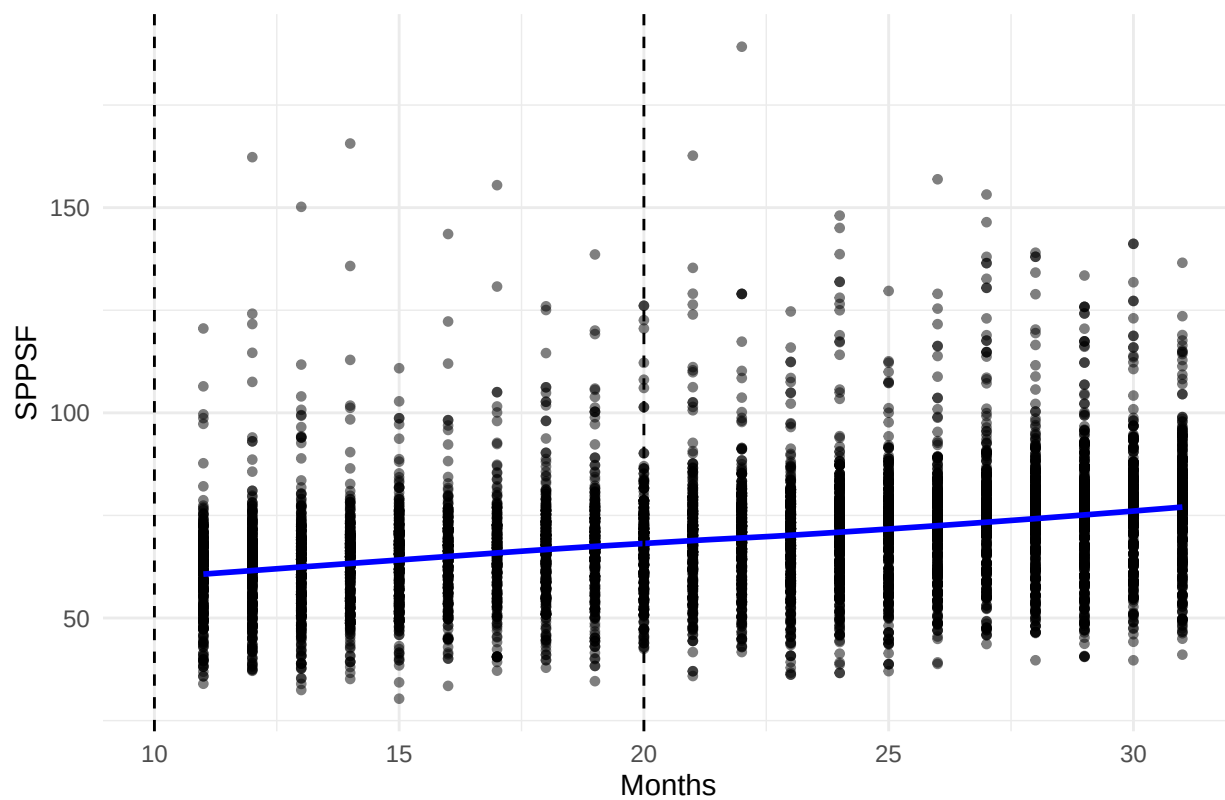
Table 2: LN_SPPSF model coefficients (95% CI)

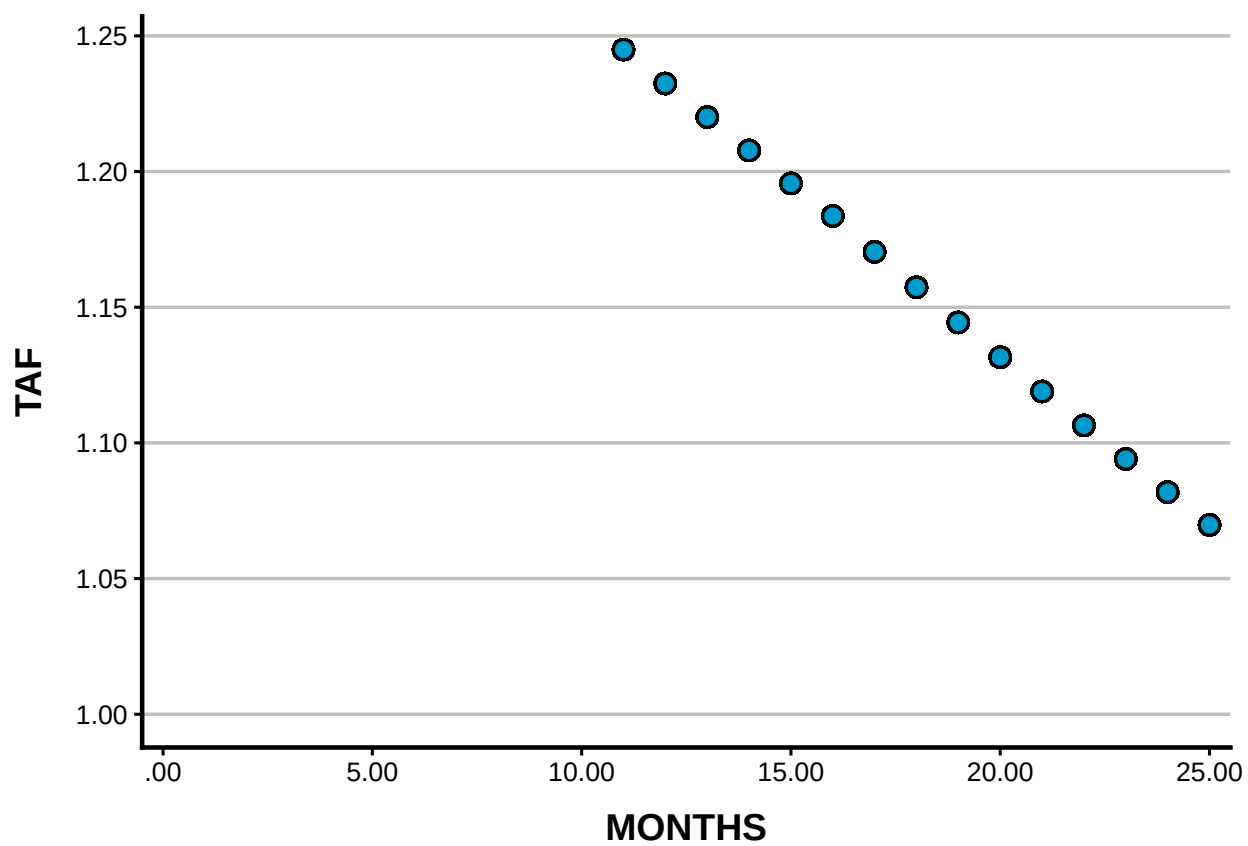
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	3.9807	128.2738	0.0310	0.0000	3.9199	4.0415

term	estimate	statistic	std_error	p_value	conf_low	conf_high
single_family1story	0.0450	2.7360	0.0165	0.0062	0.0128	0.0773
smallest	0.1425	6.2058	0.0230	0.0000	0.0975	0.1876
small	0.0702	4.4180	0.0159	0.0000	0.0391	0.1014
largest	0.0662	3.1120	0.0213	0.0019	0.0245	0.1079
good	0.2884	6.8402	0.0422	0.0000	0.2057	0.3710
fair	-0.2152	-14.8623	0.0145	0.0000	-0.2436	-0.1868
poor	-0.3967	-11.3919	0.0348	0.0000	-0.4650	-0.3284
very_poor_unsound	-0.3915	-4.2630	0.0918	0.0000	-0.5715	-0.2115
qabove_avg	0.4065	10.7101	0.0380	0.0000	0.3321	0.4809
qbelow_avg	-0.0322	-2.2527	0.0143	0.0243	-0.0603	-0.0042
yb_pre_1929	-0.1447	-8.1222	0.0178	0.0000	-0.1797	-0.1098
yb_1946_to_1960	0.0593	4.4865	0.0132	0.0000	0.0334	0.0853
slab	-0.1004	-1.8746	0.0536	0.0609	-0.2054	0.0046
ln_garagespaces	0.0512	2.8163	0.0182	0.0049	0.0156	0.0868
fireplace	0.1185	8.6842	0.0136	0.0000	0.0917	0.1452
hot_water	0.0755	3.3340	0.0227	0.0009	0.0311	0.1200
central_air	0.1039	7.1592	0.0145	0.0000	0.0754	0.1323
months_1to9	NA	NA	NA	NA	NA	NA
months_10to16	0.0102	2.1417	0.0047	0.0322	0.0009	0.0194
months_17to24	0.0113	8.1787	0.0014	0.0000	0.0086	0.0140



Months vs Predicted SPPSF – LN_SPPSF Model





Model II

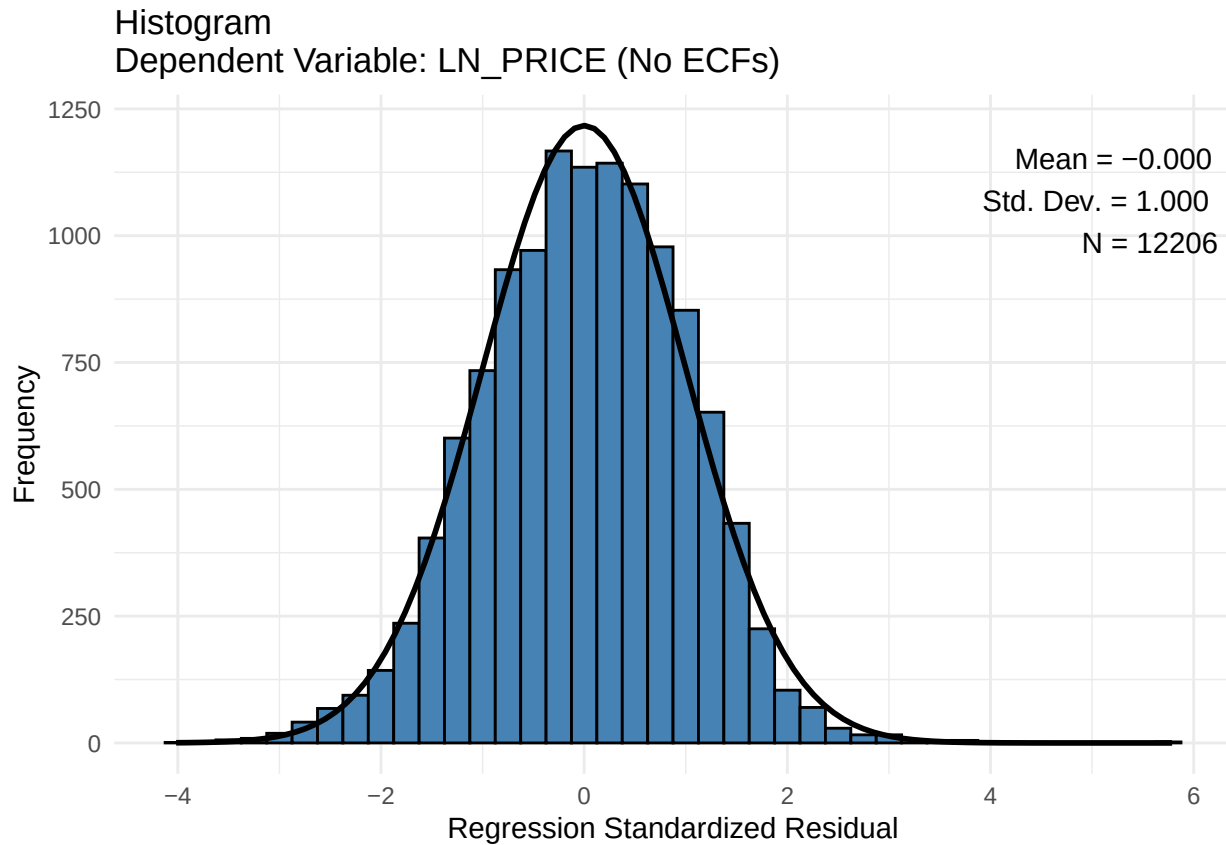


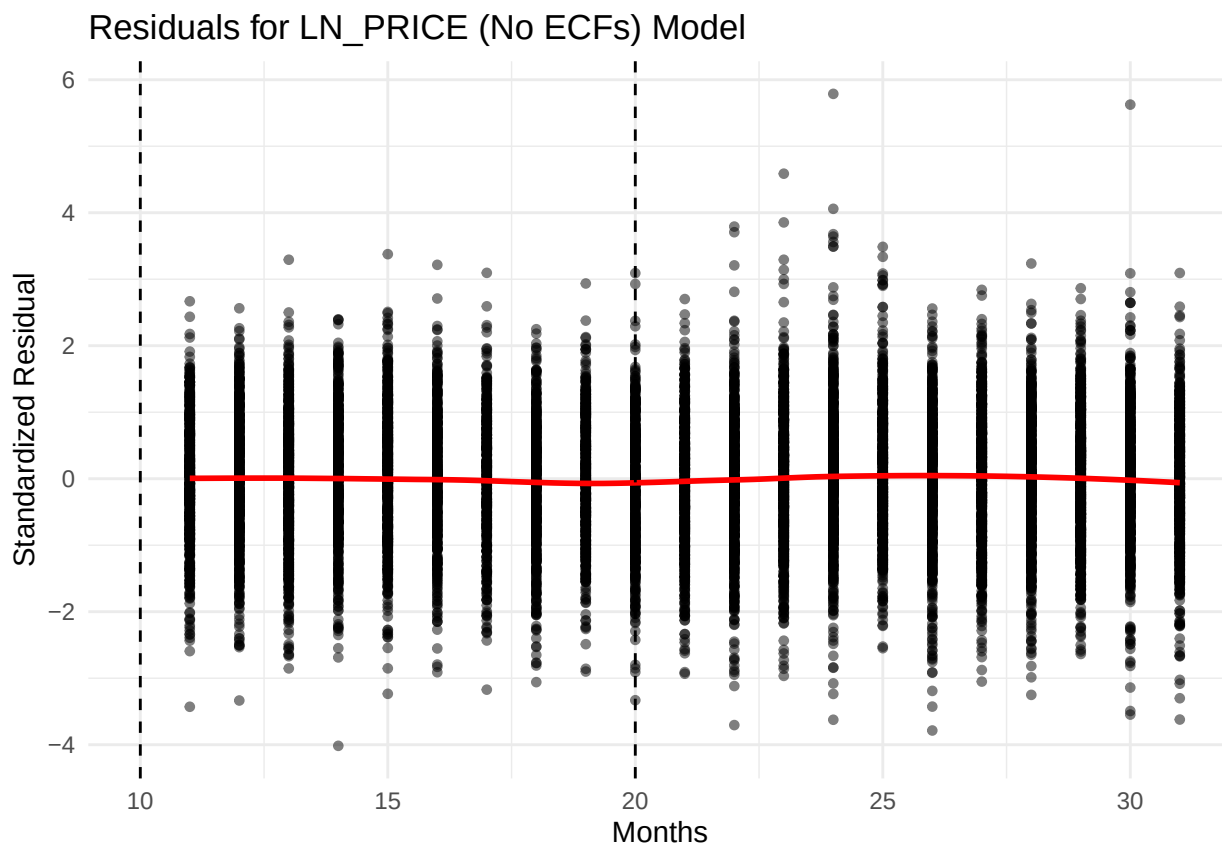
Table 3: LN_PRICE model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
12206	12182	25	0.5231643	0.3430546	0.3418142	0.5226497	0.4195776	276.5825	0	18849.55	19034.79

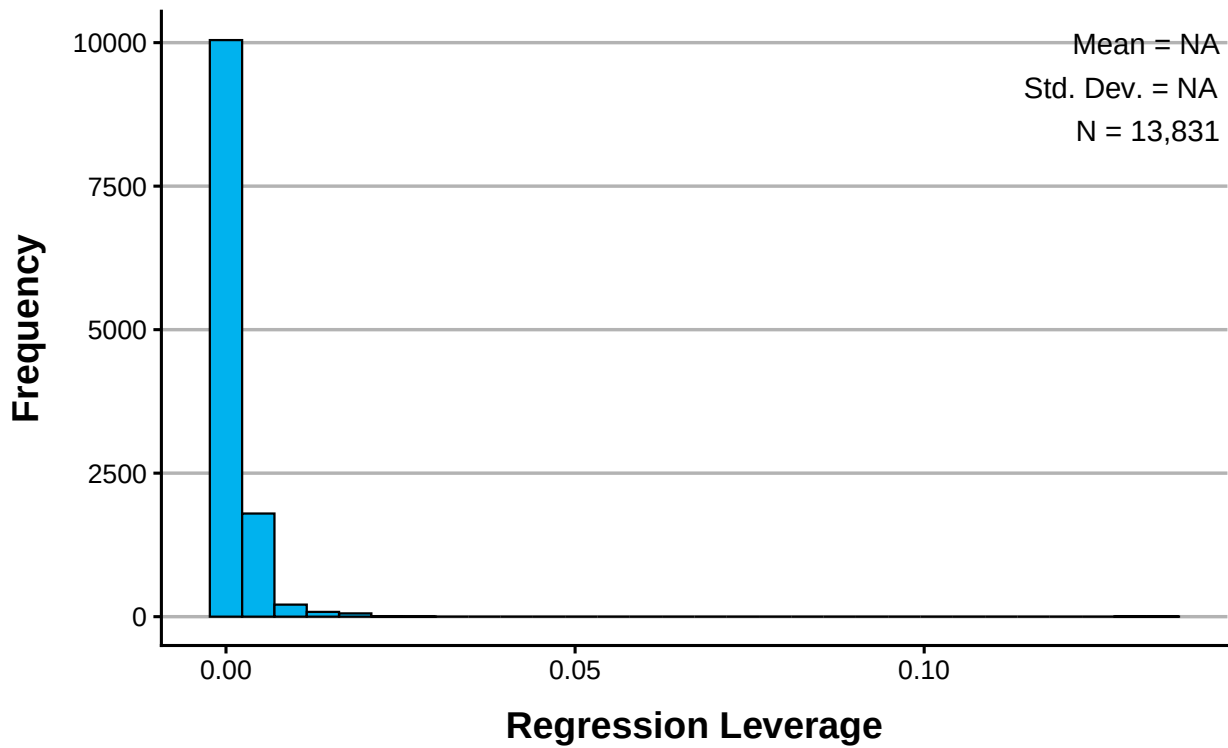
Table 4: LN_PRICE model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.9441	400.4524	0.0273	0.0000	10.8905	10.9976
ln_landratio	0.2627	11.9325	0.0220	0.0000	0.2195	0.3059
ln_base_bldgsize_ratio	0.6577	16.7486	0.0393	0.0000	0.5807	0.7346
single_family1story	0.0489	3.2975	0.0148	0.0010	0.0198	0.0779
half_duplex1story	-0.1132	-2.2430	0.0505	0.0249	-0.2121	-0.0143
large	0.0404	2.2964	0.0176	0.0217	0.0059	0.0749
largest	0.1108	3.4675	0.0320	0.0005	0.0482	0.1734
two_plus_baths	0.1677	4.5318	0.0370	0.0000	0.0952	0.2402
powder_room	0.0688	3.4067	0.0202	0.0007	0.0292	0.1084
good	0.3719	10.5367	0.0353	0.0000	0.3027	0.4410
fair	-0.2333	-19.8080	0.0118	0.0000	-0.2564	-0.2102
poor	-0.4139	-15.1160	0.0274	0.0000	-0.4676	-0.3603
very_poor_unsound	-0.3750	-5.5526	0.0675	0.0000	-0.5074	-0.2426
qabove_avg	0.3907	10.3165	0.0379	0.0000	0.3164	0.4649
qbelow_avg	-0.0478	-3.8433	0.0124	0.0001	-0.0722	-0.0234
yb_pre_1929	-0.1376	-9.5343	0.0144	0.0000	-0.1659	-0.1093
yb_1946_to_1960	0.0517	4.4600	0.0116	0.0000	0.0290	0.0744
fireplace	0.1130	9.5854	0.0118	0.0000	0.0899	0.1362

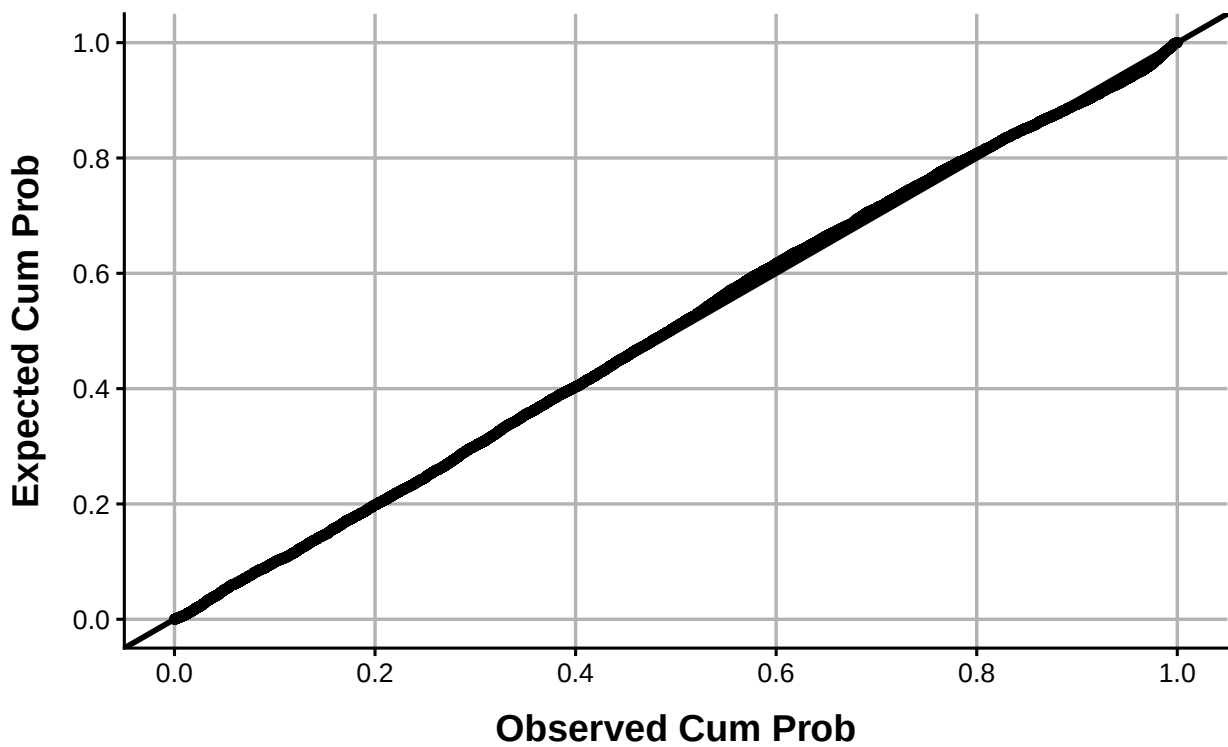
term	estimate	statistic	std_error	p_value	conf_low	conf_high
hot_water	0.0709	3.7699	0.0188	0.0002	0.0340	0.1077
elec_baseboard	-0.3169	-1.6909	0.1874	0.0909	-0.6843	0.0505
wall	-0.1662	-2.8375	0.0586	0.0046	-0.2811	-0.0514
central_air	0.1141	8.6020	0.0133	0.0000	0.0881	0.1401
months_1to9	NA	NA	NA	NA	NA	NA
months_10to16	0.0099	2.4281	0.0041	0.0152	0.0019	0.0179
months_17to24	0.0125	10.6640	0.0012	0.0000	0.0102	0.0148

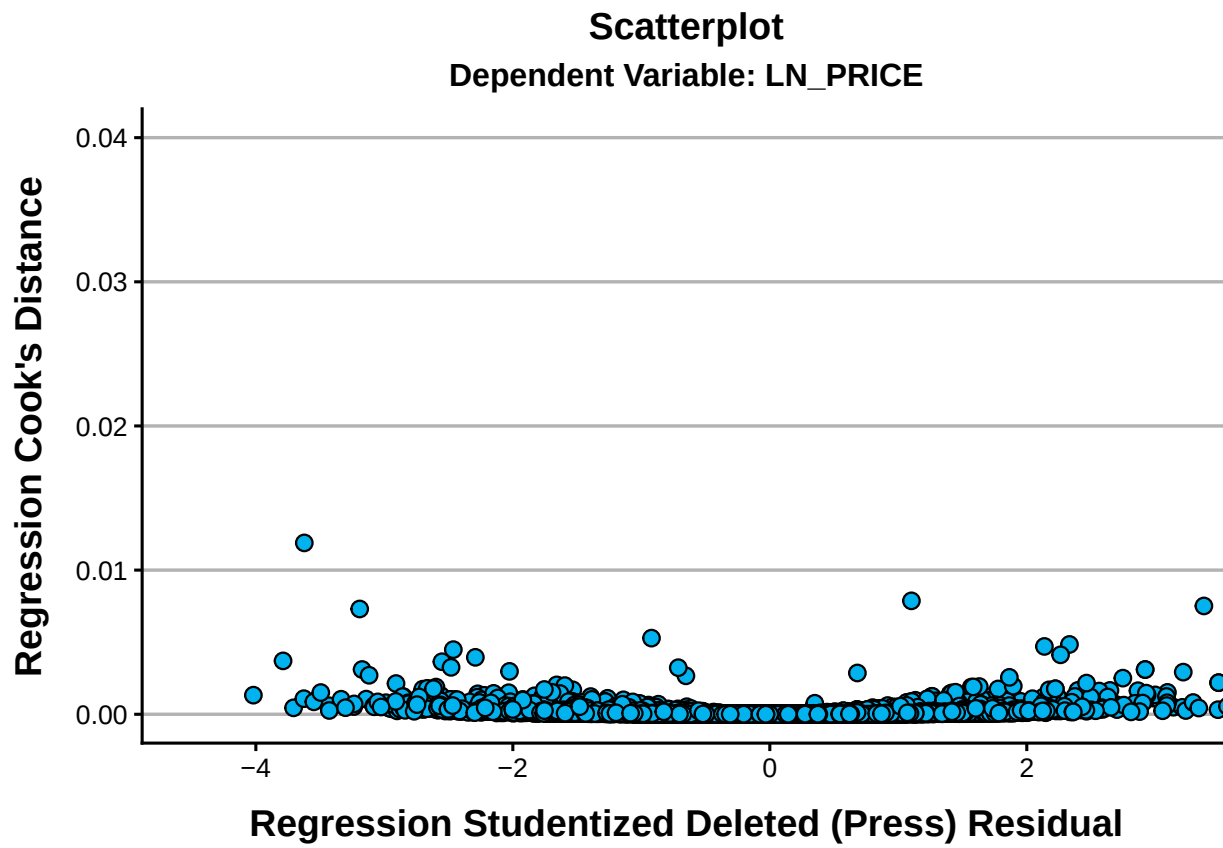
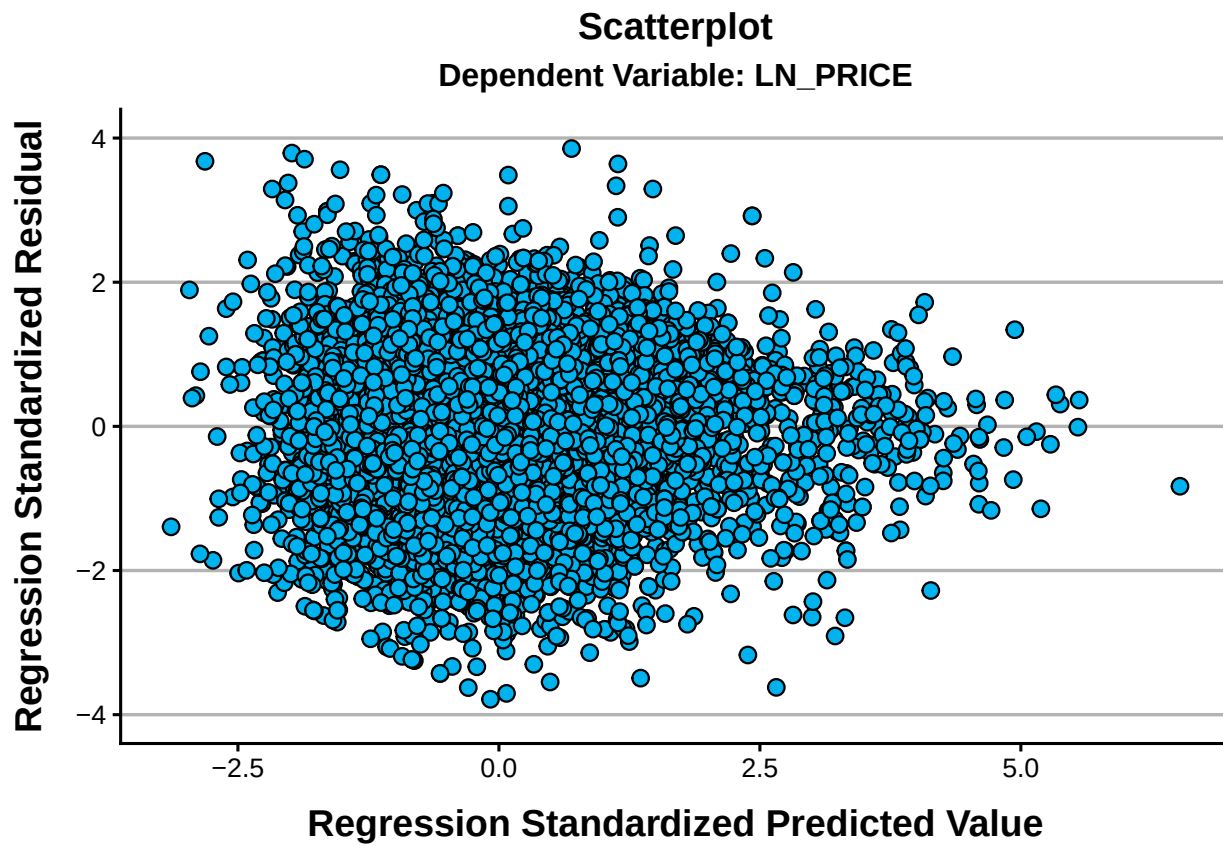


Histogram
Dependent Variable: LN_PRICE



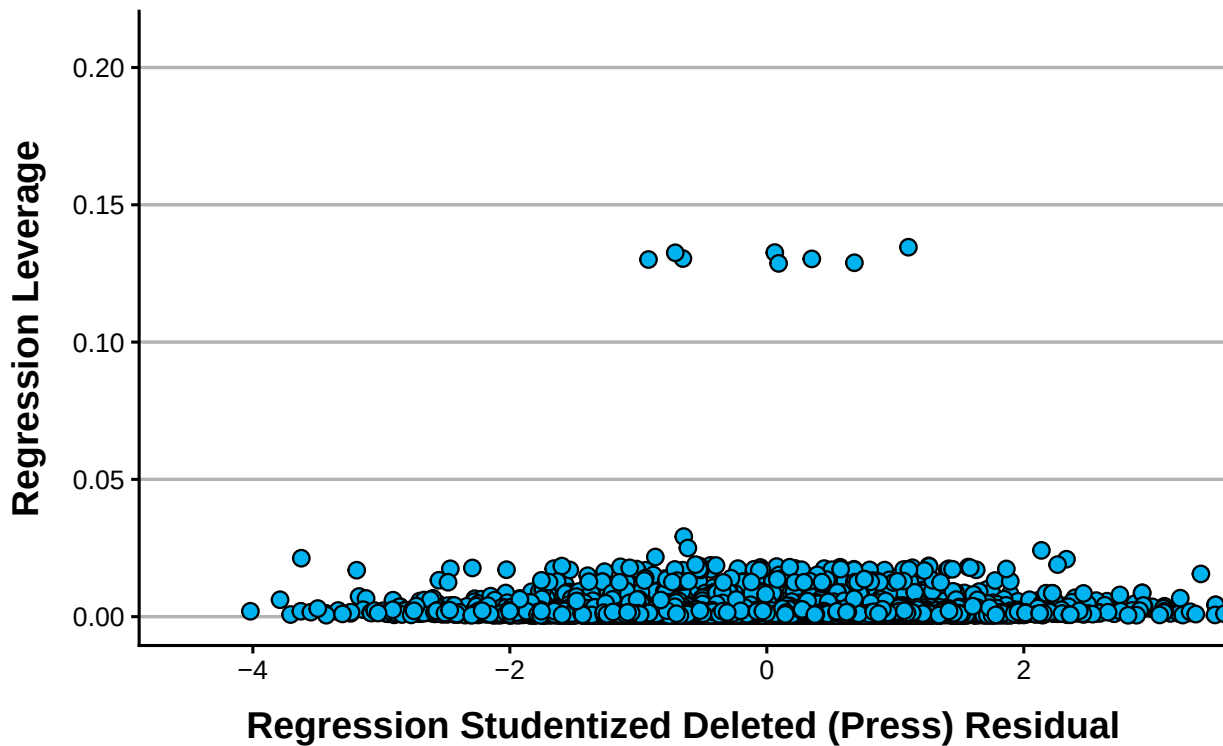
Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE





Scatterplot

Dependent Variable: LN_PRICE



Model III

```
##
## Call:
## lm(formula = new_formula, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.40603 -0.32765  0.01703  0.32435  2.84647
##
## Coefficients: (8 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    11.4723589   0.0201968  568.027 < 2e-16 ***
## ln_landratio     0.1337117   0.0264395   5.057 4.35e-07 ***
## ln_base_bldgsize_ratio 0.5662382  0.0334909  16.907 < 2e-16 ***
## half_duplex1story -0.2660616  0.0637803  -4.172 3.06e-05 ***
## largest         0.0600513   0.0223961   2.681 0.007348 **
## good            0.2531669   0.0374959   6.752 1.56e-11 ***
## fair           -0.1771005   0.0130207 -13.601 < 2e-16 ***
## poor           -0.3323436   0.0311073 -10.684 < 2e-16 ***
## very_poor_unsound -0.1856053  0.0812090  -2.286 0.022308 *
## qabove_avg      0.2369099   0.0387422   6.115 1.01e-09 ***
## qbelow_avg     -0.0775378   0.0141046  -5.497 3.97e-08 ***
## yb_pre_1929    -0.0517683   0.0181660  -2.850 0.004387 **
## yb_1946_to_1960  0.0280930   0.0125650   2.236 0.025392 *
## yb_post_1961    0.0994852   0.0394053   2.525 0.011600 *
## slab          -0.1578235   0.0494296  -3.193 0.001414 **
## ln_garagespaces  0.0329664   0.0165729   1.989 0.046715 *
```

## fireplace	0.0643019	0.0133070	4.832	1.38e-06	***
## hot_water	0.0489363	0.0205603	2.380	0.017329	*
## wall	-0.1898347	0.0795931	-2.385	0.017099	*
## central_air	0.0710698	0.0129516	5.487	4.20e-08	***
## secf_1R102	-0.3264873	0.0890725	-3.665	0.000249	***
## secf_1R104	-0.1134911	0.0411898	-2.755	0.005877	**
## secf_1R105	-0.3401980	0.0508554	-6.690	2.39e-11	***
## secf_1R106	-0.3237085	0.0649763	-4.982	6.43e-07	***
## secf_1R108	-0.4424574	0.0513121	-8.623	< 2e-16	***
## secf_1R109	-0.1825146	0.0389785	-4.682	2.88e-06	***
## secf_1R110	-0.1791047	0.0643568	-2.783	0.005398	**
## secf_1R111	-0.3371741	0.1581268	-2.132	0.033012	*
## secf_1R112	-0.1628939	0.0868808	-1.875	0.060841	.
## secf_1R113	-0.6523534	0.0830962	-7.851	4.67e-15	***
## secf_1R114	-0.3561808	0.0702441	-5.071	4.05e-07	***
## secf_1R116	0.1311693	0.0558981	2.347	0.018971	*
## secf_1R117	-0.1491215	0.0463595	-3.217	0.001302	**
## secf_1R118	-0.2802967	0.0537166	-5.218	1.85e-07	***
## secf_1R119	-0.4261578	0.0680736	-6.260	4.04e-10	***
## secf_1R120	-0.1624646	0.0800014	-2.031	0.042311	*
## secf_1R121	-0.1902360	0.0477687	-3.982	6.88e-05	***
## secf_1R122	-0.4322257	0.0558034	-7.746	1.07e-14	***
## secf_1R123	-0.3010338	0.0762113	-3.950	7.88e-05	***
## secf_1R124	-0.8702403	0.1431117	-6.081	1.25e-09	***
## secf_1R125	-0.2203142	0.0958164	-2.299	0.021511	*
## secf_1R126	0.1925636	0.0543181	3.545	0.000395	***
## secf_1R127	-0.3374314	0.0885919	-3.809	0.000141	***
## secf_1R128	-0.5456594	0.0556028	-9.814	< 2e-16	***
## secf_1R129	-0.2192383	0.0711131	-3.083	0.002057	**
## secf_1R130	-0.7949047	0.3245550	-2.449	0.014338	*
## secf_1R133	-0.3589940	0.1062464	-3.379	0.000731	***
## secf_1R135	-0.2860209	0.0591463	-4.836	1.35e-06	***
## secf_1R136	-0.3004367	0.0508112	-5.913	3.50e-09	***
## secf_1R137	-0.9383545	0.1241606	-7.558	4.56e-14	***
## secf_2R201	-0.3013779	0.0435363	-6.922	4.78e-12	***
## secf_2R202	-0.1998100	0.0765542	-2.610	0.009070	**
## secf_2R205	-0.2548415	0.0548799	-4.644	3.48e-06	***
## secf_2R208	0.4634501	0.0982741	4.716	2.45e-06	***
## secf_2R210	0.5024143	0.2282955	2.201	0.027784	*
## secf_2R211	NA	NA	NA	NA	
## secf_2R212	-0.2190373	0.0392276	-5.584	2.43e-08	***
## secf_2R215	0.2506938	0.0845818	2.964	0.003046	**
## secf_2R217	NA	NA	NA	NA	
## secf_2R218	-0.9034799	0.1352059	-6.682	2.51e-11	***
## secf_2R219	0.3559214	0.1830175	1.945	0.051841	.
## secf_2R220	-0.2374560	0.0594013	-3.997	6.46e-05	***
## secf_2R221	-0.2344614	0.0451207	-5.196	2.08e-07	***
## secf_2R222	-0.4948940	0.0516903	-9.574	< 2e-16	***
## secf_2R223	-0.2363084	0.0640568	-3.689	0.000227	***
## secf_2R224	-0.5832691	0.0629913	-9.260	< 2e-16	***
## secf_2R225	-0.2628466	0.0532098	-4.940	7.98e-07	***
## secf_2R226	-0.6929436	0.1995737	-3.472	0.000519	***
## secf_2R228	-0.6951001	0.1691736	-4.109	4.02e-05	***
## secf_2R229	-0.8894308	0.1999664	-4.448	8.79e-06	***
## secf_2R230	-0.9273614	0.2576326	-3.600	0.000321	***
## secf_3R301	-0.8691417	0.1096679	-7.925	2.58e-15	***
## secf_3R303	-0.6218150	0.0505264	-12.307	< 2e-16	***
## secf_3R304	-0.7243967	0.0929681	-7.792	7.42e-15	***
## secf_3R305	NA	NA	NA	NA	

## secf_3R306	0.1634525	0.0888966	1.839	0.065999	.
## secf_3R307	-0.4948330	0.0435790	-11.355	< 2e-16	***
## secf_3R308	-0.7600862	0.0985230	-7.715	1.36e-14	***
## secf_3R309	-0.9002360	0.1247709	-7.215	5.88e-13	***
## secf_3R310	0.2139411	0.0696946	3.070	0.002150	**
## secf_3R311	-0.4638675	0.1063907	-4.360	1.32e-05	***
## secf_3R312	-0.6928882	0.0611062	-11.339	< 2e-16	***
## secf_3R313	-0.6714225	0.0714355	-9.399	< 2e-16	***
## secf_3R314	-0.4313514	0.0710932	-6.067	1.36e-09	***
## secf_3R315	-0.4130931	0.0602169	-6.860	7.39e-12	***
## secf_3R316	-0.7845019	0.0815413	-9.621	< 2e-16	***
## secf_3R317	-0.8975167	0.1582486	-5.672	1.46e-08	***
## secf_3R318	-0.3632220	0.1997358	-1.819	0.069023	.
## secf_3R319	-0.6073225	0.0367595	-16.521	< 2e-16	***
## secf_3R320	-0.5008056	0.0450300	-11.122	< 2e-16	***
## secf_3R321	-1.0260294	0.0869627	-11.798	< 2e-16	***
## secf_3R322	-0.3438217	0.0542692	-6.335	2.49e-10	***
## secf_3R323	-0.4803805	0.0424398	-11.319	< 2e-16	***
## secf_3R324	-0.8163266	0.0922130	-8.853	< 2e-16	***
## secf_3R325	-0.5412994	0.0602753	-8.980	< 2e-16	***
## secf_3R326	-0.9506853	0.1494376	-6.362	2.10e-10	***
## secf_4R401	-0.6254305	0.0522462	-11.971	< 2e-16	***
## secf_4R402	-1.2140552	0.0893948	-13.581	< 2e-16	***
## secf_4R403	-0.5346470	0.0583232	-9.167	< 2e-16	***
## secf_4R404	-0.4564839	0.0590485	-7.731	1.20e-14	***
## secf_4R405	-0.4307173	0.0384402	-11.205	< 2e-16	***
## secf_4R406	-0.4860391	0.0379803	-12.797	< 2e-16	***
## secf_4R407	-0.6165795	0.0717039	-8.599	< 2e-16	***
## secf_4R408	-0.5977386	0.0452535	-13.209	< 2e-16	***
## secf_4R409	-0.4880333	0.0728475	-6.699	2.23e-11	***
## secf_4R410	-0.1887204	0.0350133	-5.390	7.25e-08	***
## secf_4R411	-0.6946299	0.0472356	-14.706	< 2e-16	***
## secf_4R412	-0.6659801	0.0855589	-7.784	7.91e-15	***
## secf_4R413	-0.3430793	0.0445732	-7.697	1.56e-14	***
## secf_4R416	-1.0459674	0.1581514	-6.614	3.99e-11	***
## secf_4R417	-0.2237646	0.0419992	-5.328	1.02e-07	***
## secf_4R419	-0.8793148	0.0824490	-10.665	< 2e-16	***
## secf_4R421	-0.8678170	0.0931712	-9.314	< 2e-16	***
## secf_5R501	-0.6384517	0.0646600	-9.874	< 2e-16	***
## secf_5R502	-0.6149277	0.0651036	-9.445	< 2e-16	***
## secf_5R503	0.3621306	0.1456384	2.487	0.012921	*
## secf_5R504	0.1783882	0.1071836	1.664	0.096086	.
## secf_5R505	NA	NA	NA	NA	
## secf_5R509	0.3860398	0.2003686	1.927	0.054059	.
## secf_5R510	-0.5049949	0.1829290	-2.761	0.005782	**
## secf_5R511	-0.8503373	0.4454709	-1.909	0.056317	.
## secf_5R514	0.2994297	0.1167354	2.565	0.010335	*
## secf_5R517	0.6713112	0.3167494	2.119	0.034089	*
## secf_5R521	0.4600529	0.2001686	2.298	0.021569	*
## secf_5R523	-0.6980226	0.1829041	-3.816	0.000136	***
## secf_5R524	NA	NA	NA	NA	
## secf_5R528	-0.6400209	0.2232607	-2.867	0.004159	**
## secf_5R533	-0.8959451	0.1711812	-5.234	1.70e-07	***
## secf_5R534	-0.3189163	0.1067356	-2.988	0.002817	**
## secf_5r538	-1.7136544	0.4454559	-3.847	0.000121	***
## secf_6R601	-1.2412830	0.1355268	-9.159	< 2e-16	***
## secf_6R602	-0.9274887	0.1129053	-8.215	2.46e-16	***
## secf_6R603	-1.0335828	0.1130792	-9.140	< 2e-16	***
## secf_6R604	-0.4105154	0.0805880	-5.094	3.59e-07	***

```

## secf_6R605      -0.2252838  0.1204036  -1.871  0.061371  .
## secf_6R606      -0.3726356  0.1502306  -2.480  0.013143  *
## secf_6R608       0.7313324  0.2237284   3.269  0.001084  **
## secf_6R609       1.4472774  0.4471535   3.237  0.001214  **
## secf_6R611      -0.2400108  0.0739982  -3.243  0.001186  **
## secf_6R612      -0.3191183  0.1251896  -2.549  0.010819  *
## secf_6R614       0.8778612  0.4536954   1.935  0.053036  .
## secf_6R615      -0.2803367  0.1252496  -2.238  0.025234  *
## secf_6R618      -0.7540566  0.2582391  -2.920  0.003510  **
## secf_6R620      -0.7031010  0.0547978 -12.831  < 2e-16  ***
## secf_7R703      -0.5685628  0.0861184  -6.602  4.31e-11  ***
## secf_7R704      -0.3817750  0.0373011 -10.235  < 2e-16  ***
## secf_7R705      -0.4018085  0.0265762 -15.119  < 2e-16  ***
## secf_7R706      -0.4505650  0.1092475  -4.124  3.76e-05  ***
## secf_7R707      -0.6043993  0.1350268  -4.476  7.70e-06  ***
## secf_7R708      -0.4883659  0.0578867  -8.437  < 2e-16  ***
## secf_7R709      -0.4195363  0.0517452  -8.108  5.92e-16  ***
## secf_7R710      -0.2510203  0.0526784  -4.765  1.92e-06  ***
## secf_7R711      -0.5640482  0.0495451 -11.385  < 2e-16  ***
## secf_7R712      -0.4022734  0.0924308  -4.352  1.36e-05  ***
## secf_7R713      -0.4550095  0.0713422  -6.378  1.89e-10  ***
## secf_7R714              NA              NA              NA              NA
## secf_7R715      -0.8369224  0.1996167  -4.193  2.79e-05  ***
## secf_7R716      -0.7844868  0.0989062  -7.932  2.45e-15  ***
## secf_7R717      -1.0065066  0.0721324 -13.954  < 2e-16  ***
## secf_7R718      -0.8377406  0.0659954 -12.694  < 2e-16  ***
## secf_7R719      -0.3438960  0.0357495  -9.620  < 2e-16  ***
## secf_7R720      -0.7920674  0.1689730  -4.688  2.81e-06  ***
## secf_7R721      -0.3307297  0.1587116  -2.084  0.037206  *
## secf_7R722      -0.8512563  0.1585870  -5.368  8.19e-08  ***
## secf_7R723      -0.1841800  0.0740756  -2.486  0.012925  *
## secf_7R724      -0.6704968  0.1245056  -5.385  7.44e-08  ***
## secf_CGRS6              NA              NA              NA              NA
## months_1to9              NA              NA              NA              NA
## months_17to24      0.0145004  0.0009607  15.093  < 2e-16  ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4445 on 8069 degrees of freedom
## (5601 observations deleted due to missingness)
## Multiple R-squared:  0.5093, Adjusted R-squared:  0.4996
## F-statistic: 52.34 on 160 and 8069 DF, p-value: < 2.2e-16

```

Table 5: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
8230	0.9831182	1.102229	0.9137144	1.206316	39.08367	-0.556716

Histogram

Dependent Variable: LN_PRICE (+ECFs)

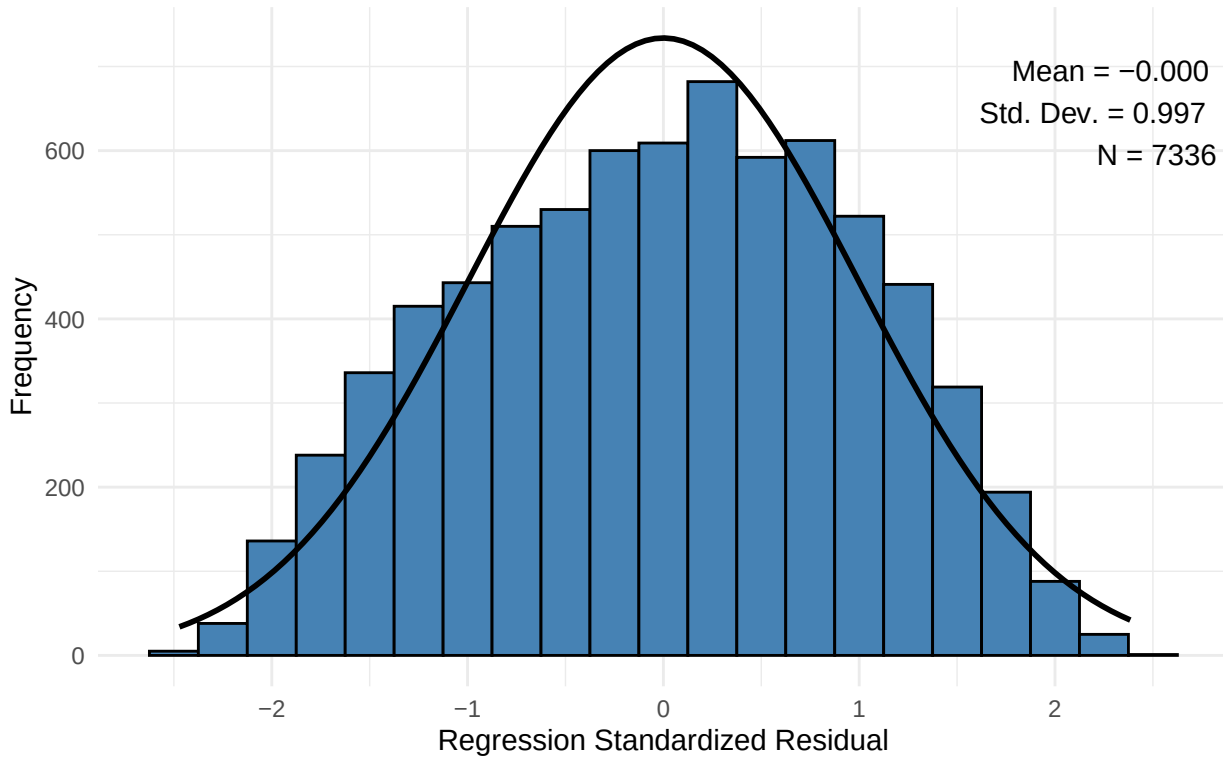


Table 6: LN_PRICE (base) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
8230	8069	169	0.444504	0.509282	0.4995516	0.4401347	0.3591518	52.33909	0	10171.43	11307.94

Table 7: LN_PRICE (+ECFs) model coefficients (95% CI)

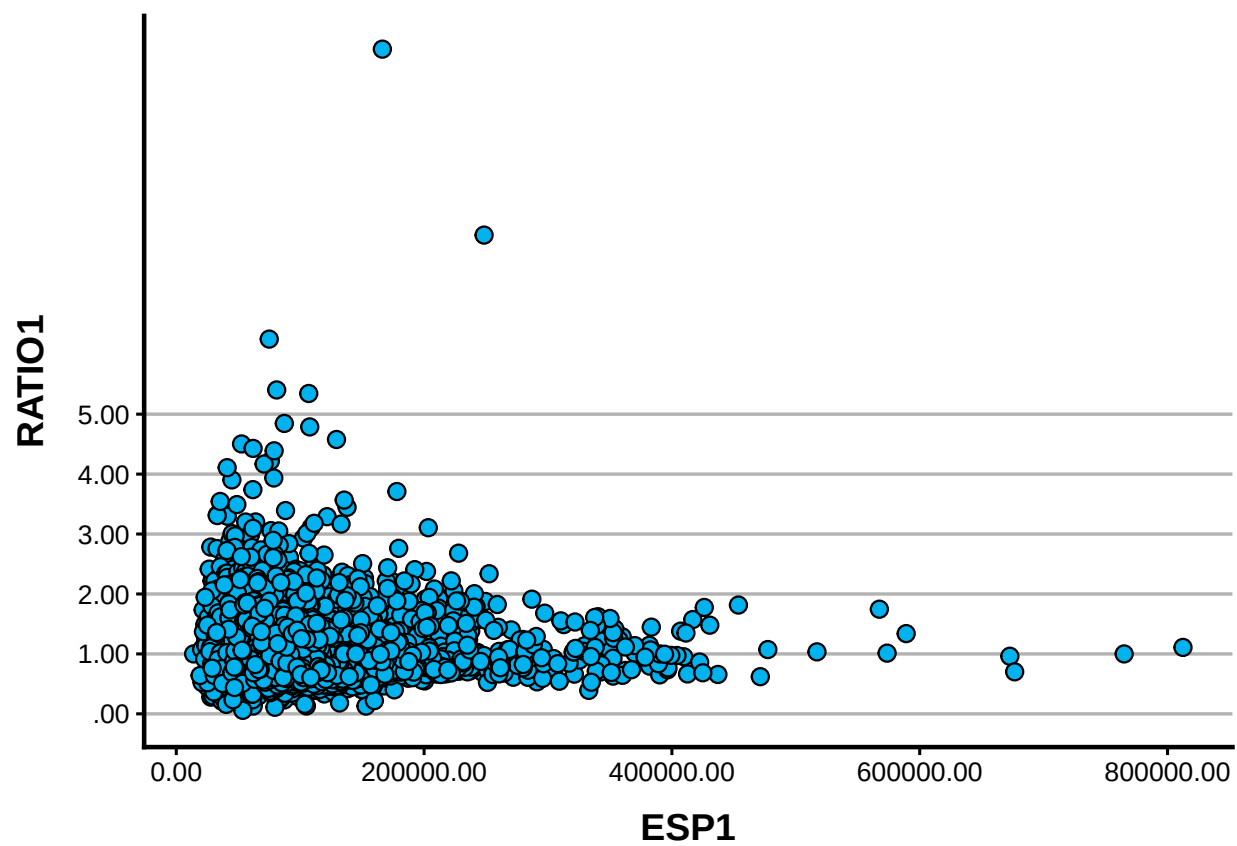
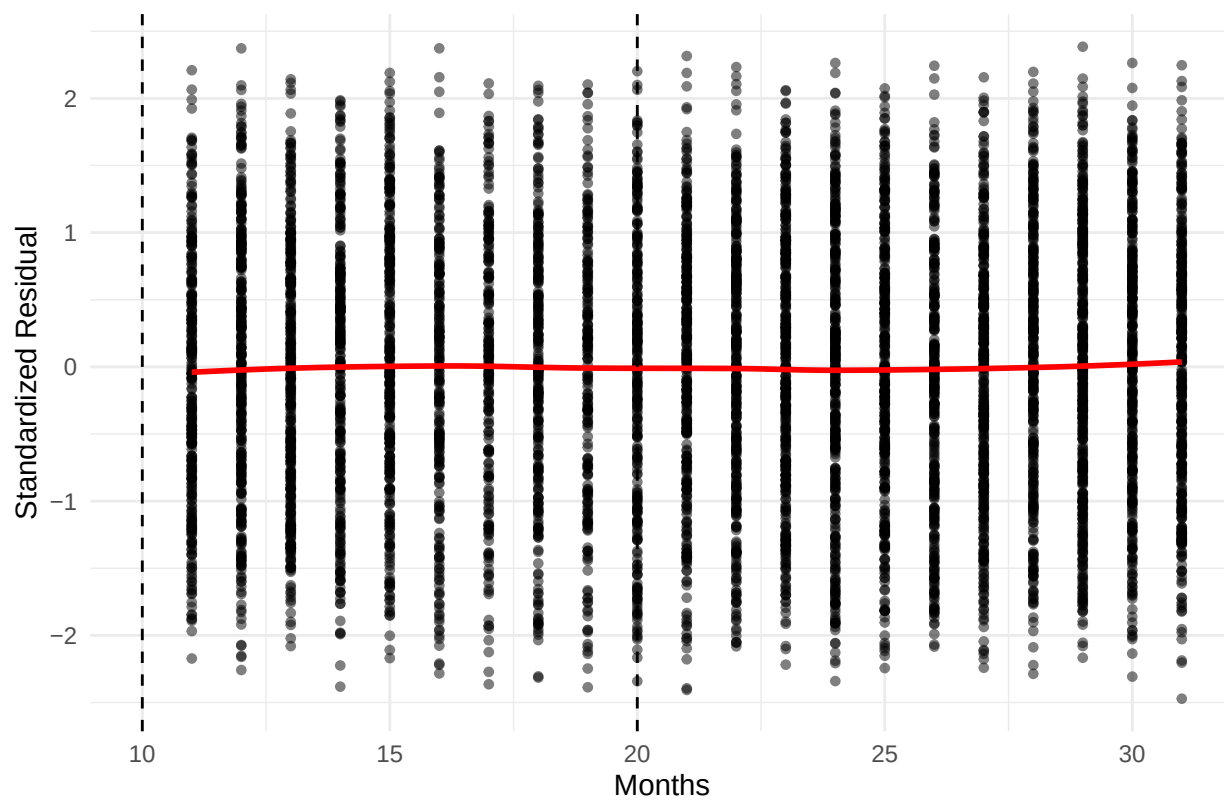
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.4724	568.0274	0.0202	0.0000	11.4328	11.5119
ln_landratio	0.1337	5.0573	0.0264	0.0000	0.0819	0.1855
ln_base_bldgsize_ratio	0.5662	16.9072	0.0335	0.0000	0.5006	0.6319
half_duplex1story	-0.2661	-4.1715	0.0638	0.0000	-0.3911	-0.1410
largest	0.0601	2.6813	0.0224	0.0073	0.0161	0.1040
good	0.2532	6.7518	0.0375	0.0000	0.1797	0.3267
fair	-0.1771	-13.6014	0.0130	0.0000	-0.2026	-0.1516
poor	-0.3323	-10.6838	0.0311	0.0000	-0.3933	-0.2714
very_poor_unsound	-0.1856	-2.2855	0.0812	0.0223	-0.3448	-0.0264
qabove_avg	0.2369	6.1150	0.0387	0.0000	0.1610	0.3129
qbelow_avg	-0.0775	-5.4973	0.0141	0.0000	-0.1052	-0.0499
yb_pre_1929	-0.0518	-2.8497	0.0182	0.0044	-0.0874	-0.0162
yb_1946_to_1960	0.0281	2.2358	0.0126	0.0254	0.0035	0.0527
yb_post_1961	0.0995	2.5247	0.0394	0.0116	0.0222	0.1767
slab	-0.1578	-3.1929	0.0494	0.0014	-0.2547	-0.0609
ln_garagespaces	0.0330	1.9892	0.0166	0.0467	0.0005	0.0655
fireplace	0.0643	4.8322	0.0133	0.0000	0.0382	0.0904
hot_water	0.0489	2.3801	0.0206	0.0173	0.0086	0.0892
wall	-0.1898	-2.3851	0.0796	0.0171	-0.3459	-0.0338
central_air	0.0711	5.4873	0.0130	0.0000	0.0457	0.0965

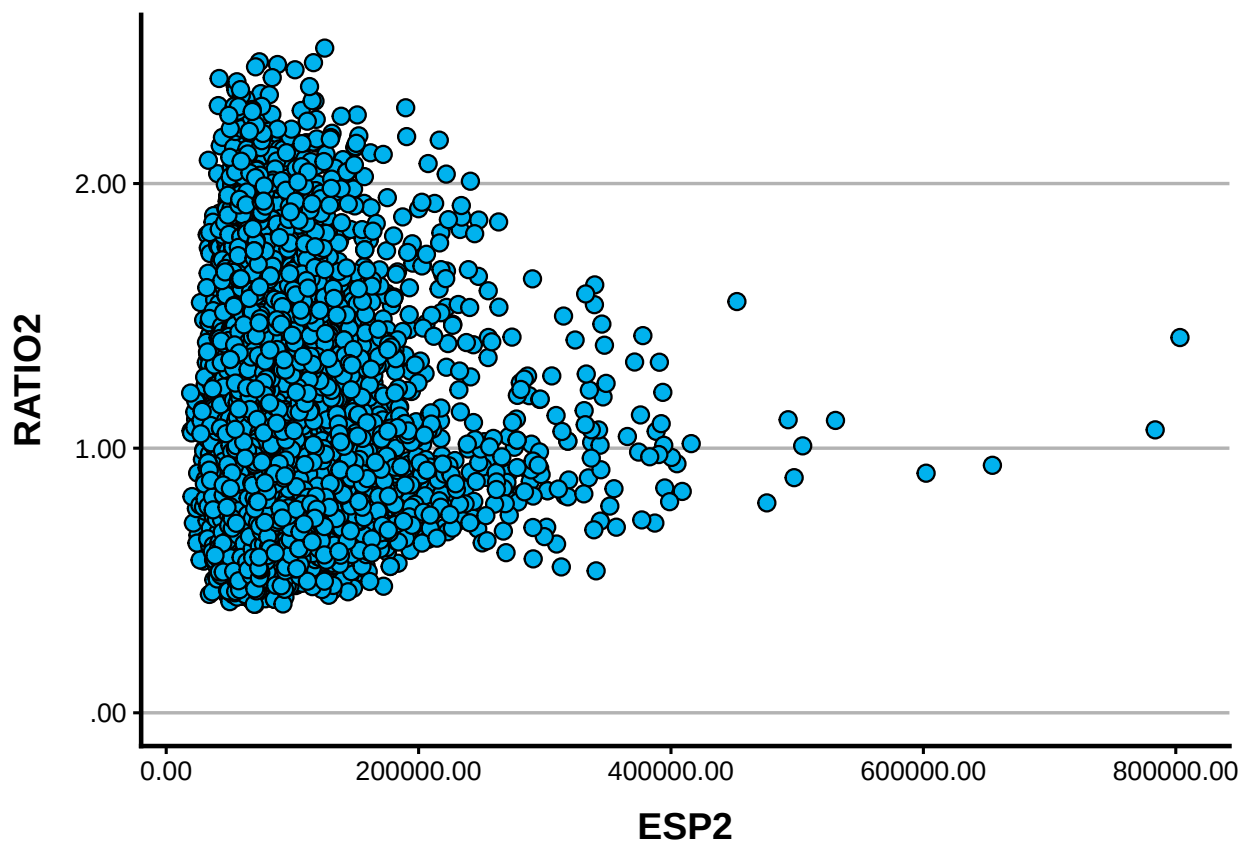
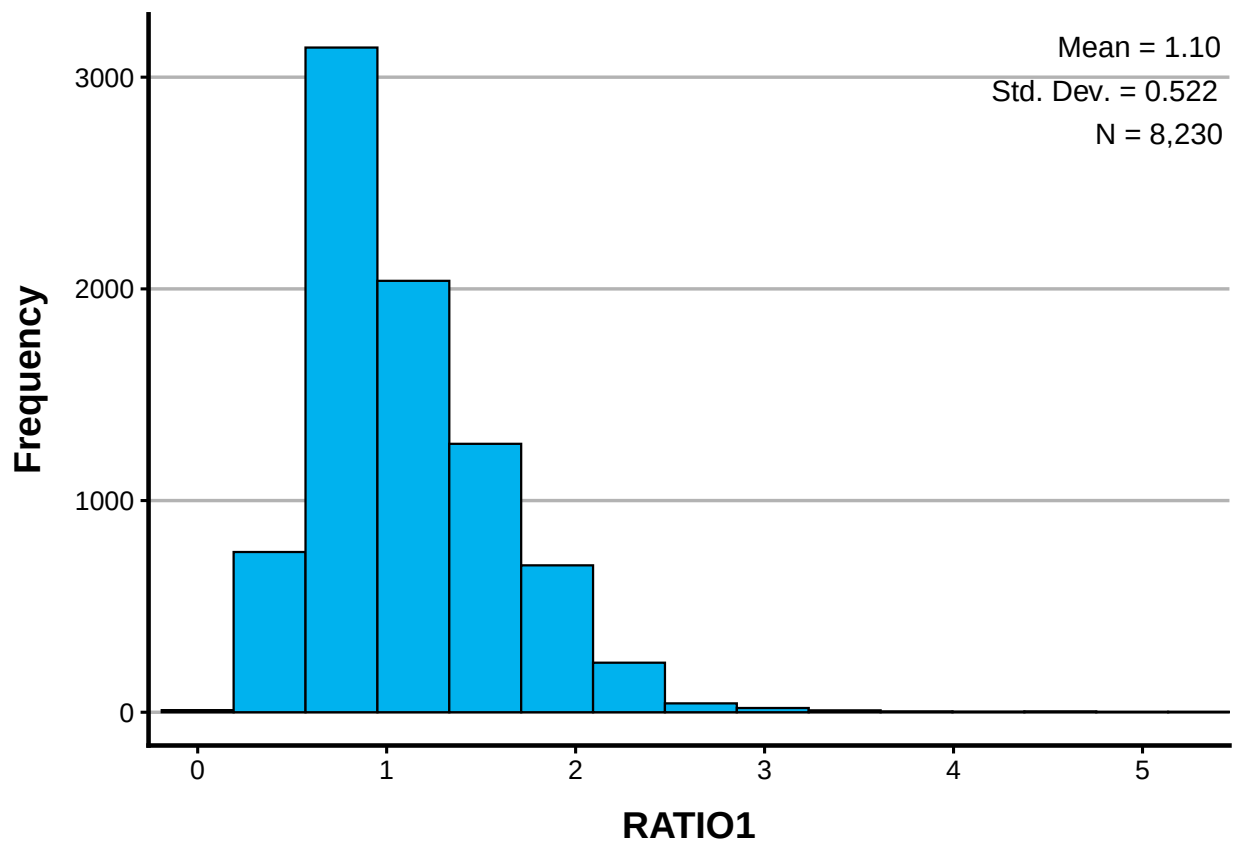
term	estimate	statistic	std_error	p_value	conf_low	conf_high
secf_1R102	-0.3265	-3.6654	0.0891	0.0002	-0.5011	-0.1519
secf_1R104	-0.1135	-2.7553	0.0412	0.0059	-0.1942	-0.0327
secf_1R105	-0.3402	-6.6895	0.0509	0.0000	-0.4399	-0.2405
secf_1R106	-0.3237	-4.9819	0.0650	0.0000	-0.4511	-0.1963
secf_1R108	-0.4425	-8.6229	0.0513	0.0000	-0.5430	-0.3419
secf_1R109	-0.1825	-4.6824	0.0390	0.0000	-0.2589	-0.1061
secf_1R110	-0.1791	-2.7830	0.0644	0.0054	-0.3053	-0.0529
secf_1R111	-0.3372	-2.1323	0.1581	0.0330	-0.6471	-0.0272
secf_1R112	-0.1629	-1.8749	0.0869	0.0608	-0.3332	0.0074
secf_1R113	-0.6524	-7.8506	0.0831	0.0000	-0.8152	-0.4895
secf_1R114	-0.3562	-5.0706	0.0702	0.0000	-0.4939	-0.2185
secf_1R116	0.1312	2.3466	0.0559	0.0190	0.0216	0.2407
secf_1R117	-0.1491	-3.2166	0.0464	0.0013	-0.2400	-0.0582
secf_1R118	-0.2803	-5.2181	0.0537	0.0000	-0.3856	-0.1750
secf_1R119	-0.4262	-6.2603	0.0681	0.0000	-0.5596	-0.2927
secf_1R120	-0.1625	-2.0308	0.0800	0.0423	-0.3193	-0.0056
secf_1R121	-0.1902	-3.9824	0.0478	0.0001	-0.2839	-0.0966
secf_1R122	-0.4322	-7.7455	0.0558	0.0000	-0.5416	-0.3228
secf_1R123	-0.3010	-3.9500	0.0762	0.0001	-0.4504	-0.1516
secf_1R124	-0.8702	-6.0808	0.1431	0.0000	-1.1508	-0.5897
secf_1R125	-0.2203	-2.2993	0.0958	0.0215	-0.4081	-0.0325
secf_1R126	0.1926	3.5451	0.0543	0.0004	0.0861	0.2990
secf_1R127	-0.3374	-3.8088	0.0886	0.0001	-0.5111	-0.1638
secf_1R128	-0.5457	-9.8135	0.0556	0.0000	-0.6547	-0.4367
secf_1R129	-0.2192	-3.0830	0.0711	0.0021	-0.3586	-0.0798
secf_1R130	-0.7949	-2.4492	0.3246	0.0143	-1.4311	-0.1587
secf_1R133	-0.3590	-3.3789	0.1062	0.0007	-0.5673	-0.1507
secf_1R135	-0.2860	-4.8358	0.0591	0.0000	-0.4020	-0.1701
secf_1R136	-0.3004	-5.9128	0.0508	0.0000	-0.4000	-0.2008
secf_1R137	-0.9384	-7.5576	0.1242	0.0000	-1.1817	-0.6950
secf_2R201	-0.3014	-6.9224	0.0435	0.0000	-0.3867	-0.2160
secf_2R202	-0.1998	-2.6100	0.0766	0.0091	-0.3499	-0.0497
secf_2R205	-0.2548	-4.6436	0.0549	0.0000	-0.3624	-0.1473
secf_2R208	0.4635	4.7159	0.0983	0.0000	0.2708	0.6561
secf_2R210	0.5024	2.2007	0.2283	0.0278	0.0549	0.9499
secf_2R211	NA	NA	NA	NA	NA	NA
secf_2R212	-0.2190	-5.5838	0.0392	0.0000	-0.2959	-0.1421
secf_2R215	0.2507	2.9639	0.0846	0.0030	0.0849	0.4165
secf_2R217	NA	NA	NA	NA	NA	NA
secf_2R218	-0.9035	-6.6823	0.1352	0.0000	-1.1685	-0.6384
secf_2R219	0.3559	1.9447	0.1830	0.0518	-0.0028	0.7147
secf_2R220	-0.2375	-3.9975	0.0594	0.0001	-0.3539	-0.1210
secf_2R221	-0.2345	-5.1963	0.0451	0.0000	-0.3229	-0.1460
secf_2R222	-0.4949	-9.5742	0.0517	0.0000	-0.5962	-0.3936
secf_2R223	-0.2363	-3.6890	0.0641	0.0002	-0.3619	-0.1107
secf_2R224	-0.5833	-9.2595	0.0630	0.0000	-0.7067	-0.4598
secf_2R225	-0.2628	-4.9398	0.0532	0.0000	-0.3672	-0.1585
secf_2R226	-0.6929	-3.4721	0.1996	0.0005	-1.0842	-0.3017
secf_2R228	-0.6951	-4.1088	0.1692	0.0000	-1.0267	-0.3635
secf_2R229	-0.8894	-4.4479	0.2000	0.0000	-1.2814	-0.4974
secf_2R230	-0.9274	-3.5995	0.2576	0.0003	-1.4324	-0.4223
secf_3R301	-0.8691	-7.9252	0.1097	0.0000	-1.0841	-0.6542
secf_3R303	-0.6218	-12.3067	0.0505	0.0000	-0.7209	-0.5228
secf_3R304	-0.7244	-7.7919	0.0930	0.0000	-0.9066	-0.5422
secf_3R305	NA	NA	NA	NA	NA	NA
secf_3R306	0.1635	1.8387	0.0889	0.0660	-0.0108	0.3377
secf_3R307	-0.4948	-11.3548	0.0436	0.0000	-0.5803	-0.4094

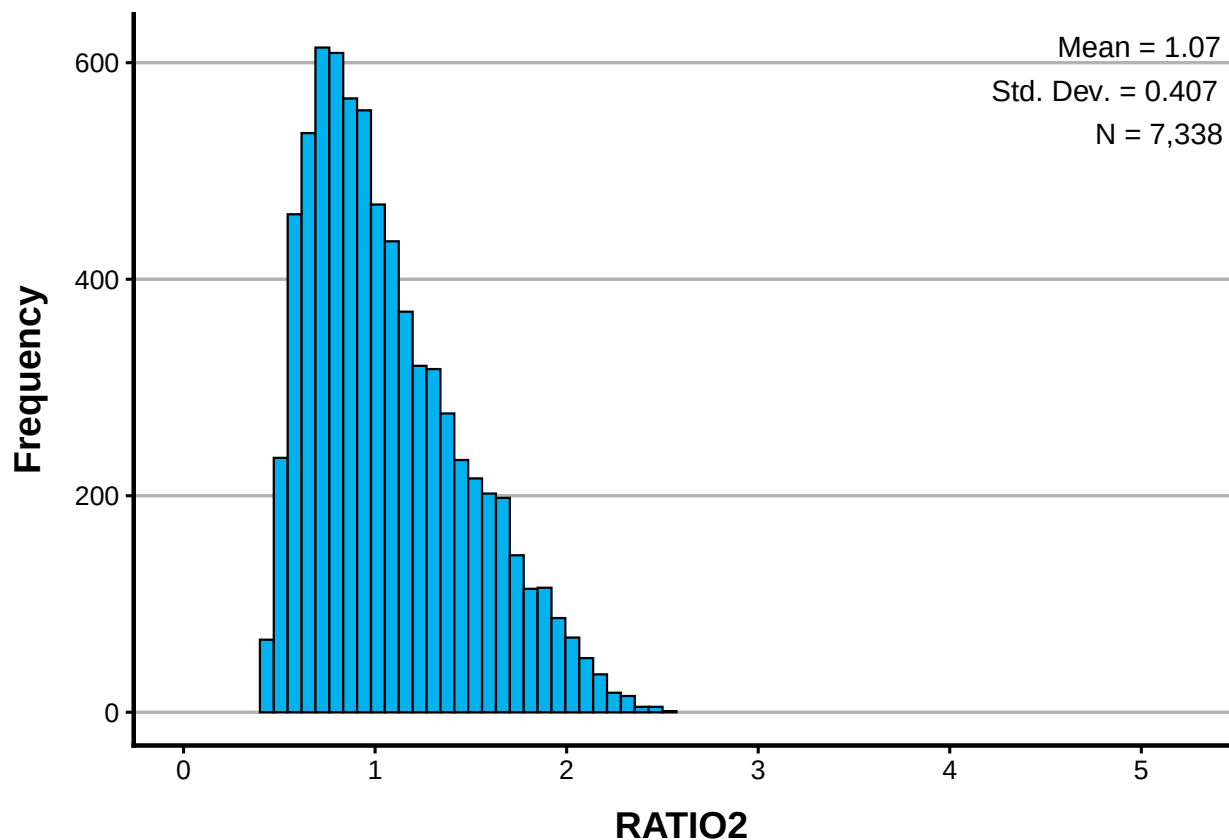
term	estimate	statistic	std_error	p_value	conf_low	conf_high
secf_3R308	-0.7601	-7.7148	0.0985	0.0000	-0.9532	-0.5670
secf_3R309	-0.9002	-7.2151	0.1248	0.0000	-1.1448	-0.6557
secf_3R310	0.2139	3.0697	0.0697	0.0021	0.0773	0.3506
secf_3R311	-0.4639	-4.3600	0.1064	0.0000	-0.6724	-0.2553
secf_3R312	-0.6929	-11.3391	0.0611	0.0000	-0.8127	-0.5731
secf_3R313	-0.6714	-9.3990	0.0714	0.0000	-0.8115	-0.5314
secf_3R314	-0.4314	-6.0674	0.0711	0.0000	-0.5707	-0.2920
secf_3R315	-0.4131	-6.8601	0.0602	0.0000	-0.5311	-0.2951
secf_3R316	-0.7845	-9.6209	0.0815	0.0000	-0.9443	-0.6247
secf_3R317	-0.8975	-5.6716	0.1582	0.0000	-1.2077	-0.5873
secf_3R318	-0.3632	-1.8185	0.1997	0.0690	-0.7548	0.0283
secf_3R319	-0.6073	-16.5215	0.0368	0.0000	-0.6794	-0.5353
secf_3R320	-0.5008	-11.1216	0.0450	0.0000	-0.5891	-0.4125
secf_3R321	-1.0260	-11.7985	0.0870	0.0000	-1.1965	-0.8556
secf_3R322	-0.3438	-6.3355	0.0543	0.0000	-0.4502	-0.2374
secf_3R323	-0.4804	-11.3191	0.0424	0.0000	-0.5636	-0.3972
secf_3R324	-0.8163	-8.8526	0.0922	0.0000	-0.9971	-0.6356
secf_3R325	-0.5413	-8.9805	0.0603	0.0000	-0.6595	-0.4231
secf_3R326	-0.9507	-6.3618	0.1494	0.0000	-1.2436	-0.6577
secf_4R401	-0.6254	-11.9708	0.0522	0.0000	-0.7278	-0.5230
secf_4R402	-1.2141	-13.5808	0.0894	0.0000	-1.3893	-1.0388
secf_4R403	-0.5346	-9.1670	0.0583	0.0000	-0.6490	-0.4203
secf_4R404	-0.4565	-7.7307	0.0590	0.0000	-0.5722	-0.3407
secf_4R405	-0.4307	-11.2049	0.0384	0.0000	-0.5061	-0.3554
secf_4R406	-0.4860	-12.7971	0.0380	0.0000	-0.5605	-0.4116
secf_4R407	-0.6166	-8.5990	0.0717	0.0000	-0.7571	-0.4760
secf_4R408	-0.5977	-13.2087	0.0453	0.0000	-0.6864	-0.5090
secf_4R409	-0.4880	-6.6994	0.0728	0.0000	-0.6308	-0.3452
secf_4R410	-0.1887	-5.3900	0.0350	0.0000	-0.2574	-0.1201
secf_4R411	-0.6946	-14.7057	0.0472	0.0000	-0.7872	-0.6020
secf_4R412	-0.6660	-7.7839	0.0856	0.0000	-0.8337	-0.4983
secf_4R413	-0.3431	-7.6970	0.0446	0.0000	-0.4305	-0.2557
secf_4R416	-1.0460	-6.6137	0.1582	0.0000	-1.3560	-0.7359
secf_4R417	-0.2238	-5.3278	0.0420	0.0000	-0.3061	-0.1414
secf_4R419	-0.8793	-10.6650	0.0824	0.0000	-1.0409	-0.7177
secf_4R421	-0.8678	-9.3142	0.0932	0.0000	-1.0505	-0.6852
secf_5R501	-0.6385	-9.8740	0.0647	0.0000	-0.7652	-0.5117
secf_5R502	-0.6149	-9.4454	0.0651	0.0000	-0.7425	-0.4873
secf_5R503	0.3621	2.4865	0.1456	0.0129	0.0766	0.6476
secf_5R504	0.1784	1.6643	0.1072	0.0961	-0.0317	0.3885
secf_5R505	NA	NA	NA	NA	NA	NA
secf_5R509	0.3860	1.9266	0.2004	0.0541	-0.0067	0.7788
secf_5R510	-0.5050	-2.7606	0.1829	0.0058	-0.8636	-0.1464
secf_5R511	-0.8503	-1.9089	0.4455	0.0563	-1.7236	0.0229
secf_5R514	0.2994	2.5650	0.1167	0.0103	0.0706	0.5283
secf_5R517	0.6713	2.1194	0.3167	0.0341	0.0504	1.2922
secf_5R521	0.4601	2.2983	0.2002	0.0216	0.0677	0.8524
secf_5R523	-0.6980	-3.8163	0.1829	0.0001	-1.0566	-0.3395
secf_5R524	NA	NA	NA	NA	NA	NA
secf_5R528	-0.6400	-2.8667	0.2233	0.0042	-1.0777	-0.2024
secf_5R533	-0.8959	-5.2339	0.1712	0.0000	-1.2315	-0.5604
secf_5R534	-0.3189	-2.9879	0.1067	0.0028	-0.5281	-0.1097
secf_5r538	-1.7137	-3.8470	0.4455	0.0001	-2.5869	-0.8404
secf_6R601	-1.2413	-9.1590	0.1355	0.0000	-1.5070	-0.9756
secf_6R602	-0.9275	-8.2147	0.1129	0.0000	-1.1488	-0.7062
secf_6R603	-1.0336	-9.1403	0.1131	0.0000	-1.2552	-0.8119
secf_6R604	-0.4105	-5.0940	0.0806	0.0000	-0.5685	-0.2525

term	estimate	statistic	std_error	p_value	conf_low	conf_high
secf_6R605	-0.2253	-1.8711	0.1204	0.0614	-0.4613	0.0107
secf_6R606	-0.3726	-2.4804	0.1502	0.0131	-0.6671	-0.0781
secf_6R608	0.7313	3.2688	0.2237	0.0011	0.2928	1.1699
secf_6R609	1.4473	3.2366	0.4472	0.0012	0.5707	2.3238
secf_6R611	-0.2400	-3.2435	0.0740	0.0012	-0.3851	-0.0950
secf_6R612	-0.3191	-2.5491	0.1252	0.0108	-0.5645	-0.0737
secf_6R614	0.8779	1.9349	0.4537	0.0530	-0.0115	1.7672
secf_6R615	-0.2803	-2.2382	0.1252	0.0252	-0.5259	-0.0348
secf_6R618	-0.7541	-2.9200	0.2582	0.0035	-1.2603	-0.2478
secf_6R620	-0.7031	-12.8308	0.0548	0.0000	-0.8105	-0.5957
secf_7R703	-0.5686	-6.6021	0.0861	0.0000	-0.7374	-0.3997
secf_7R704	-0.3818	-10.2350	0.0373	0.0000	-0.4549	-0.3087
secf_7R705	-0.4018	-15.1191	0.0266	0.0000	-0.4539	-0.3497
secf_7R706	-0.4506	-4.1243	0.1092	0.0000	-0.6647	-0.2364
secf_7R707	-0.6044	-4.4761	0.1350	0.0000	-0.8691	-0.3397
secf_7R708	-0.4884	-8.4366	0.0579	0.0000	-0.6018	-0.3749
secf_7R709	-0.4195	-8.1077	0.0517	0.0000	-0.5210	-0.3181
secf_7R710	-0.2510	-4.7651	0.0527	0.0000	-0.3543	-0.1478
secf_7R711	-0.5640	-11.3845	0.0495	0.0000	-0.6612	-0.4669
secf_7R712	-0.4023	-4.3522	0.0924	0.0000	-0.5835	-0.2211
secf_7R713	-0.4550	-6.3778	0.0713	0.0000	-0.5949	-0.3152
secf_7R714	NA	NA	NA	NA	NA	NA
secf_7R715	-0.8369	-4.1926	0.1996	0.0000	-1.2282	-0.4456
secf_7R716	-0.7845	-7.9316	0.0989	0.0000	-0.9784	-0.5906
secf_7R717	-1.0065	-13.9536	0.0721	0.0000	-1.1479	-0.8651
secf_7R718	-0.8377	-12.6939	0.0660	0.0000	-0.9671	-0.7084
secf_7R719	-0.3439	-9.6196	0.0357	0.0000	-0.4140	-0.2738
secf_7R720	-0.7921	-4.6875	0.1690	0.0000	-1.1233	-0.4608
secf_7R721	-0.3307	-2.0838	0.1587	0.0372	-0.6418	-0.0196
secf_7R722	-0.8513	-5.3678	0.1586	0.0000	-1.1621	-0.5404
secf_7R723	-0.1842	-2.4864	0.0741	0.0129	-0.3294	-0.0390
secf_7R724	-0.6705	-5.3853	0.1245	0.0000	-0.9146	-0.4264
secf_CGRS6	NA	NA	NA	NA	NA	NA
months_1to9	NA	NA	NA	NA	NA	NA
months_17to24	0.0145	15.0929	0.0010	0.0000	0.0126	0.0164

Residuals for LN_PRICE (+ECFs) Model







Model IV

Table 8: LN_TASP model assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
7338	1.119403	1.221614	1.06248	1.149775	33.51992	-0.5329354

Table 9: Alternative LN_TASP model assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
7338	1.120663	1.221501	1.062264	1.149904	33.46039	-0.5323286

Table 10: LN_TASP model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
7338	7191	164	0.3760505	0.574614	0.5659773	0.3722648	0.3105348	66.53183	0	6618.256	7639.578

Table 11: LN_TASP model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.7056	670.6866	0.0175	0.0000	11.6714	11.7399
ln_landratio	0.1335	5.5712	0.0240	0.0000	0.0865	0.1805

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ln_base_bldgsize_ratio	0.5993	19.8401	0.0302	0.0000	0.5401	0.6586
half_duplex1story	-0.2796	-4.2416	0.0659	0.0000	-0.4088	-0.1504
largest	0.0533	2.6086	0.0204	0.0091	0.0132	0.0933
good	0.2410	7.0474	0.0342	0.0000	0.1739	0.3080
fair	-0.1704	-14.3863	0.0118	0.0000	-0.1936	-0.1471
poor	-0.3086	-9.8294	0.0314	0.0000	-0.3702	-0.2471
very_poor_unsound	-0.2138	-1.7614	0.1214	0.0782	-0.4518	0.0242
qabove_avg	0.2169	5.9013	0.0367	0.0000	0.1448	0.2889
qbelow_avg	-0.0785	-6.1803	0.0127	0.0000	-0.1034	-0.0536
yb_pre_1929	-0.0479	-2.9077	0.0165	0.0037	-0.0801	-0.0156
yb_1946_to_1960	0.0260	2.3541	0.0110	0.0186	0.0043	0.0476
yb_post_1961	0.1407	3.8014	0.0370	0.0001	0.0682	0.2133
slab	-0.2939	-5.7454	0.0512	0.0000	-0.3942	-0.1937
fireplace	0.0623	5.2385	0.0119	0.0000	0.0390	0.0857
hot_water	0.0397	2.0606	0.0192	0.0394	0.0019	0.0774
wall	-0.1580	-1.8003	0.0877	0.0719	-0.3300	0.0140
central_air	0.0767	6.7231	0.0114	0.0000	0.0544	0.0991
secf_1R102	-0.2658	-2.8694	0.0926	0.0041	-0.4474	-0.0842
secf_1R104	-0.1210	-3.4231	0.0353	0.0006	-0.1902	-0.0517
secf_1R105	-0.3405	-7.8087	0.0436	0.0000	-0.4260	-0.2550
secf_1R106	-0.3167	-5.4582	0.0580	0.0000	-0.4305	-0.2030
secf_1R108	-0.4170	-9.0709	0.0460	0.0000	-0.5071	-0.3269
secf_1R109	-0.1946	-5.8139	0.0335	0.0000	-0.2602	-0.1290
secf_1R110	-0.1537	-2.7551	0.0558	0.0059	-0.2630	-0.0443
secf_1R111	-0.4076	-2.1589	0.1888	0.0309	-0.7776	-0.0375
secf_1R113	-0.6657	-6.9785	0.0954	0.0000	-0.8527	-0.4787
secf_1R114	-0.3564	-4.8285	0.0738	0.0000	-0.5011	-0.2117
secf_1R116	0.1141	2.3944	0.0476	0.0167	0.0207	0.2074
secf_1R117	-0.1644	-4.1386	0.0397	0.0000	-0.2423	-0.0865
secf_1R118	-0.2407	-5.0858	0.0473	0.0000	-0.3335	-0.1479
secf_1R119	-0.4601	-6.8621	0.0671	0.0000	-0.5916	-0.3287
secf_1R120	-0.2113	-2.7078	0.0780	0.0068	-0.3642	-0.0583
secf_1R121	-0.1904	-4.6246	0.0412	0.0000	-0.2711	-0.1097
secf_1R122	-0.4530	-9.1409	0.0496	0.0000	-0.5502	-0.3559
secf_1R123	-0.2389	-3.4388	0.0695	0.0006	-0.3751	-0.1027
secf_1R124	-0.8711	-4.5824	0.1901	0.0000	-1.2438	-0.4985
secf_1R125	-0.2766	-2.1897	0.1263	0.0286	-0.5242	-0.0290
secf_1R126	0.1967	4.1931	0.0469	0.0000	0.1047	0.2886
secf_1R127	-0.3311	-3.4829	0.0951	0.0005	-0.5175	-0.1448
secf_1R128	-0.5587	-11.5509	0.0484	0.0000	-0.6535	-0.4639
secf_1R129	-0.2738	-4.2810	0.0639	0.0000	-0.3991	-0.1484
secf_1R130	NA	NA	NA	NA	NA	NA
secf_1R133	-0.3478	-2.4302	0.1431	0.0151	-0.6283	-0.0672
secf_1R135	-0.3211	-6.2308	0.0515	0.0000	-0.4222	-0.2201
secf_1R136	-0.3040	-6.9736	0.0436	0.0000	-0.3895	-0.2186
secf_1R137	-1.0052	-7.0380	0.1428	0.0000	-1.2852	-0.7253
secf_2R201	-0.3112	-8.3368	0.0373	0.0000	-0.3844	-0.2381
secf_2R202	-0.2767	-3.7584	0.0736	0.0002	-0.4210	-0.1324
secf_2R203	-0.0874	-2.3905	0.0366	0.0169	-0.1591	-0.0157
secf_2R205	-0.1791	-3.6387	0.0492	0.0003	-0.2757	-0.0826
secf_2R208	0.4749	5.2899	0.0898	0.0000	0.2989	0.6508
secf_2R210	0.4555	1.6826	0.2707	0.0925	-0.0752	0.9862
secf_2R211	NA	NA	NA	NA	NA	NA
secf_2R212	-0.2315	-6.8612	0.0337	0.0000	-0.2977	-0.1654
secf_2R213	-0.0842	-2.0757	0.0405	0.0380	-0.1636	-0.0047
secf_2R215	0.2226	3.0157	0.0738	0.0026	0.0779	0.3673
secf_2R216	0.3358	1.9247	0.1745	0.0543	-0.0062	0.6778

term	estimate	statistic	std_error	p_value	conf_low	conf_high
secf_2R217	NA	NA	NA	NA	NA	NA
secf_2R218	-0.9771	-5.7191	0.1709	0.0000	-1.3121	-0.6422
secf_2R219	0.3956	2.0789	0.1903	0.0377	0.0226	0.7687
secf_2R220	-0.1946	-3.6596	0.0532	0.0003	-0.2988	-0.0903
secf_2R221	-0.2809	-7.1322	0.0394	0.0000	-0.3581	-0.2037
secf_2R222	-0.5338	-11.9731	0.0446	0.0000	-0.6212	-0.4464
secf_2R223	-0.1997	-3.4929	0.0572	0.0005	-0.3118	-0.0876
secf_2R224	-0.6366	-11.2372	0.0566	0.0000	-0.7476	-0.5255
secf_2R225	-0.2542	-5.2978	0.0480	0.0000	-0.3483	-0.1602
secf_2R226	NA	NA	NA	NA	NA	NA
secf_2R228	-0.6697	-1.7774	0.3768	0.0755	-1.4083	0.0689
secf_2R229	-0.7394	-1.9620	0.3769	0.0498	-1.4781	-0.0006
secf_2R230	NA	NA	NA	NA	NA	NA
secf_3R301	-0.7989	-3.6662	0.2179	0.0002	-1.2261	-0.3717
secf_3R303	-0.6179	-14.1276	0.0437	0.0000	-0.7036	-0.5322
secf_3R304	-0.7650	-6.3725	0.1201	0.0000	-1.0004	-0.5297
secf_3R305	NA	NA	NA	NA	NA	NA
secf_3R306	0.2690	2.2308	0.1206	0.0257	0.0326	0.5053
secf_3R307	-0.5041	-13.5095	0.0373	0.0000	-0.5773	-0.4310
secf_3R308	-0.9383	-7.8243	0.1199	0.0000	-1.1734	-0.7032
secf_3R309	-0.7613	-4.9303	0.1544	0.0000	-1.0640	-0.4586
secf_3R310	0.2939	4.4618	0.0659	0.0000	0.1648	0.4230
secf_3R311	-0.4120	-3.5883	0.1148	0.0003	-0.6371	-0.1869
secf_3R312	-0.7226	-12.3134	0.0587	0.0000	-0.8376	-0.6076
secf_3R313	-0.6778	-8.9977	0.0753	0.0000	-0.8255	-0.5301
secf_3R314	-0.3714	-5.4626	0.0680	0.0000	-0.5046	-0.2381
secf_3R315	-0.3747	-6.6893	0.0560	0.0000	-0.4845	-0.2649
secf_3R316	-0.7857	-8.4992	0.0925	0.0000	-0.9670	-0.6045
secf_3R317	-0.8883	-3.3330	0.2665	0.0009	-1.4107	-0.3658
secf_3R318	-0.3775	-1.7325	0.2179	0.0832	-0.8046	0.0496
secf_3R319	-0.6477	-20.2029	0.0321	0.0000	-0.7106	-0.5849
secf_3R320	-0.5240	-13.5294	0.0387	0.0000	-0.5999	-0.4481
secf_3R321	-1.1030	-9.6505	0.1143	0.0000	-1.3270	-0.8789
secf_3R322	-0.3481	-7.4669	0.0466	0.0000	-0.4395	-0.2567
secf_3R323	-0.5033	-13.7605	0.0366	0.0000	-0.5749	-0.4316
secf_3R324	-0.8688	-8.5473	0.1016	0.0000	-1.0681	-0.6696
secf_3R325	-0.5869	-11.2581	0.0521	0.0000	-0.6891	-0.4847
secf_3R326	-0.9825	-5.2059	0.1887	0.0000	-1.3524	-0.6125
secf_4R401	-0.7006	-15.2550	0.0459	0.0000	-0.7906	-0.6106
secf_4R402	-1.3659	-14.5801	0.0937	0.0000	-1.5495	-1.1822
secf_4R403	-0.5750	-10.9808	0.0524	0.0000	-0.6777	-0.4724
secf_4R404	-0.4305	-7.9747	0.0540	0.0000	-0.5363	-0.3246
secf_4R405	-0.4443	-13.4631	0.0330	0.0000	-0.5090	-0.3796
secf_4R406	-0.5132	-15.6280	0.0328	0.0000	-0.5775	-0.4488
secf_4R407	-0.6088	-8.9854	0.0678	0.0000	-0.7417	-0.4760
secf_4R408	-0.6027	-15.4997	0.0389	0.0000	-0.6790	-0.5265
secf_4R409	-0.4829	-7.3096	0.0661	0.0000	-0.6124	-0.3534
secf_4R410	-0.2175	-7.1472	0.0304	0.0000	-0.2772	-0.1578
secf_4R411	-0.7061	-16.4262	0.0430	0.0000	-0.7903	-0.6218
secf_4R412	-0.6218	-6.8640	0.0906	0.0000	-0.7994	-0.4442
secf_4R413	-0.3534	-9.0991	0.0388	0.0000	-0.4295	-0.2772
secf_4R416	-0.9324	-2.4747	0.3768	0.0134	-1.6710	-0.1938
secf_4R417	-0.2339	-6.4738	0.0361	0.0000	-0.3047	-0.1631
secf_4R419	-0.9554	-12.2399	0.0781	0.0000	-1.1085	-0.8024
secf_4R421	-0.9041	-11.1210	0.0813	0.0000	-1.0635	-0.7447
secf_5R501	-0.7937	-11.3687	0.0698	0.0000	-0.9306	-0.6569
secf_5R502	-0.6653	-8.9199	0.0746	0.0000	-0.8115	-0.5191

term	estimate	statistic	std_error	p_value	conf_low	conf_high
secf_5R503	0.3743	1.9548	0.1915	0.0506	-0.0010	0.7496
secf_5R504	0.2588	2.4613	0.1052	0.0139	0.0527	0.4649
secf_5R505	NA	NA	NA	NA	NA	NA
secf_5R509	0.4339	1.9892	0.2181	0.0467	0.0063	0.8615
secf_5R510	NA	NA	NA	NA	NA	NA
secf_5R511	NA	NA	NA	NA	NA	NA
secf_5R514	0.3547	2.9382	0.1207	0.0033	0.1181	0.5914
secf_5R517	0.6284	2.3401	0.2686	0.0193	0.1020	1.1549
secf_5R523	-0.7596	-2.8457	0.2669	0.0044	-1.2828	-0.2363
secf_5R524	NA	NA	NA	NA	NA	NA
secf_5R528	-0.6097	-2.2859	0.2667	0.0223	-1.1326	-0.0868
secf_5R533	-0.8278	-3.0787	0.2689	0.0021	-1.3549	-0.3007
secf_5R534	-0.3308	-1.7476	0.1893	0.0806	-0.7018	0.0403
secf_5r538	NA	NA	NA	NA	NA	NA
secf_6R601	-1.2729	-7.5203	0.1693	0.0000	-1.6047	-0.9411
secf_6R602	-0.9657	-5.1075	0.1891	0.0000	-1.3364	-0.5951
secf_6R603	-1.2365	-9.1937	0.1345	0.0000	-1.5001	-0.9728
secf_6R604	-0.3949	-4.5590	0.0866	0.0000	-0.5646	-0.2251
secf_6R605	-0.2494	-2.0740	0.1202	0.0381	-0.4850	-0.0137
secf_6R606	-0.5119	-2.3465	0.2181	0.0190	-0.9395	-0.0843
secf_6R609	NA	NA	NA	NA	NA	NA
secf_6R611	-0.2352	-3.2204	0.0730	0.0013	-0.3783	-0.0920
secf_6R612	-0.2765	-1.7858	0.1548	0.0742	-0.5800	0.0270
secf_6R614	NA	NA	NA	NA	NA	NA
secf_6R618	NA	NA	NA	NA	NA	NA
secf_6R620	-0.7311	-14.8411	0.0493	0.0000	-0.8277	-0.6346
secf_7R703	-0.5319	-5.8730	0.0906	0.0000	-0.7094	-0.3544
secf_7R704	-0.3841	-11.9509	0.0321	0.0000	-0.4471	-0.3211
secf_7R705	-0.4059	-17.4063	0.0233	0.0000	-0.4516	-0.3602
secf_7R706	-0.5246	-4.5816	0.1145	0.0000	-0.7491	-0.3001
secf_7R707	-0.6245	-2.8691	0.2176	0.0041	-1.0511	-0.1978
secf_7R708	-0.5518	-10.6869	0.0516	0.0000	-0.6530	-0.4506
secf_7R709	-0.4084	-9.0076	0.0453	0.0000	-0.4973	-0.3195
secf_7R710	-0.2245	-4.9083	0.0457	0.0000	-0.3142	-0.1348
secf_7R711	-0.5892	-13.8484	0.0425	0.0000	-0.6726	-0.5058
secf_7R712	-0.4479	-4.8355	0.0926	0.0000	-0.6295	-0.2664
secf_7R713	-0.5062	-7.3124	0.0692	0.0000	-0.6420	-0.3705
secf_7R714	NA	NA	NA	NA	NA	NA
secf_7R715	NA	NA	NA	NA	NA	NA
secf_7R716	-0.6920	-6.0298	0.1148	0.0000	-0.9170	-0.4671
secf_7R717	-1.1151	-15.4713	0.0721	0.0000	-1.2564	-0.9738
secf_7R718	-0.8841	-14.4739	0.0611	0.0000	-1.0039	-0.7644
secf_7R719	-0.3479	-11.2520	0.0309	0.0000	-0.4085	-0.2873
secf_7R720	-0.7148	-2.6828	0.2664	0.0073	-1.2371	-0.1925
secf_7R721	-0.4395	-1.6498	0.2664	0.0990	-0.9617	0.0827
secf_7R722	-0.8124	-3.0473	0.2666	0.0023	-1.3349	-0.2898
secf_7R724	-0.6310	-4.0854	0.1544	0.0000	-0.9337	-0.3282
secf_CGRS6	NA	NA	NA	NA	NA	NA

Table 12: Alternative LN_TASP model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
7338	7191	165	0.3759513	0.5748385	0.5662063	0.3721666	0.3104387	66.59295	0	6614.384	7635.705

Table 13: Alternative LN_TASP model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	9.4783	80.1418	0.1183	0.0000	9.2465	9.7102
ln_landratio	0.1387	5.7798	0.0240	0.0000	0.0917	0.1857
lnsbldsfint	0.3481	19.4450	0.0179	0.0000	0.3130	0.3832
half_duplex1story	-0.1403	-2.1310	0.0658	0.0331	-0.2693	-0.0112
largest	0.0746	3.6426	0.0205	0.0003	0.0344	0.1147
two_plus_baths	0.0614	2.0594	0.0298	0.0395	0.0030	0.1198
good	0.2405	7.0267	0.0342	0.0000	0.1734	0.3076
fair	-0.1703	-14.3815	0.0118	0.0000	-0.1935	-0.1471
poor	-0.3075	-9.7927	0.0314	0.0000	-0.3690	-0.2459
very_poor_unsound	-0.2121	-1.7472	0.1214	0.0806	-0.4500	0.0259
qabove_avg	0.2248	5.9785	0.0376	0.0000	0.1511	0.2986
qbelow_avg	-0.0802	-6.2911	0.0128	0.0000	-0.1052	-0.0552
yb_pre_1929	-0.0490	-2.9797	0.0164	0.0029	-0.0812	-0.0168
yb_1946_to_1960	0.0209	1.8922	0.0110	0.0585	-0.0008	0.0425
yb_post_1961	0.1384	3.7389	0.0370	0.0002	0.0659	0.2110
slab	-0.3006	-5.9646	0.0504	0.0000	-0.3994	-0.2018
garage	NA	NA	NA	NA	NA	NA
fireplace	0.0616	5.1771	0.0119	0.0000	0.0383	0.0850
hot_water	0.0417	2.1673	0.0192	0.0302	0.0040	0.0795
central_air	0.0763	6.6881	0.0114	0.0000	0.0539	0.0987
secf_1R102	-0.2634	-2.8437	0.0926	0.0045	-0.4449	-0.0818
secf_1R104	-0.1236	-3.4974	0.0353	0.0005	-0.1928	-0.0543
secf_1R105	-0.3395	-7.7884	0.0436	0.0000	-0.4250	-0.2541
secf_1R106	-0.3216	-5.5463	0.0580	0.0000	-0.4353	-0.2079
secf_1R108	-0.4201	-9.1445	0.0459	0.0000	-0.5101	-0.3300
secf_1R109	-0.1947	-5.8204	0.0334	0.0000	-0.2602	-0.1291
secf_1R110	-0.1561	-2.7995	0.0557	0.0051	-0.2653	-0.0468
secf_1R111	-0.4104	-2.1746	0.1887	0.0297	-0.7804	-0.0405
secf_1R113	-0.6683	-7.0080	0.0954	0.0000	-0.8553	-0.4814
secf_1R114	-0.3569	-4.8363	0.0738	0.0000	-0.5016	-0.2123
secf_1R116	0.1124	2.3526	0.0478	0.0187	0.0187	0.2060
secf_1R117	-0.1696	-4.2690	0.0397	0.0000	-0.2474	-0.0917
secf_1R118	-0.2403	-5.0790	0.0473	0.0000	-0.3331	-0.1476
secf_1R119	-0.4615	-6.8868	0.0670	0.0000	-0.5929	-0.3302
secf_1R120	-0.2144	-2.7489	0.0780	0.0060	-0.3673	-0.0615
secf_1R121	-0.1897	-4.6083	0.0412	0.0000	-0.2704	-0.1090
secf_1R122	-0.4526	-9.1313	0.0496	0.0000	-0.5498	-0.3554
secf_1R123	-0.2368	-3.4094	0.0695	0.0007	-0.3730	-0.1007
secf_1R124	-0.8981	-4.7535	0.1889	0.0000	-1.2684	-0.5277
secf_1R125	-0.2788	-2.2077	0.1263	0.0273	-0.5263	-0.0312
secf_1R126	0.1984	4.2301	0.0469	0.0000	0.1064	0.2903
secf_1R127	-0.3227	-3.3923	0.0951	0.0007	-0.5092	-0.1362
secf_1R128	-0.5586	-11.5500	0.0484	0.0000	-0.6534	-0.4638
secf_1R129	-0.2848	-4.4758	0.0636	0.0000	-0.4095	-0.1601
secf_1R130	NA	NA	NA	NA	NA	NA
secf_1R133	-0.3408	-2.3822	0.1431	0.0172	-0.6213	-0.0604
secf_1R135	-0.3204	-6.2168	0.0515	0.0000	-0.4214	-0.2193
secf_1R136	-0.3052	-7.0023	0.0436	0.0000	-0.3907	-0.2198
secf_1R137	-1.0037	-7.0293	0.1428	0.0000	-1.2837	-0.7238
secf_2R201	-0.3123	-8.3689	0.0373	0.0000	-0.3855	-0.2392
secf_2R202	-0.2766	-3.7574	0.0736	0.0002	-0.4209	-0.1323
secf_2R203	-0.0869	-2.3765	0.0366	0.0175	-0.1586	-0.0152
secf_2R205	-0.1781	-3.6220	0.0492	0.0003	-0.2745	-0.0817
secf_2R208	0.4725	5.2649	0.0898	0.0000	0.2966	0.6485
secf_2R210	0.5966	2.2084	0.2701	0.0272	0.0670	1.1262

term	estimate	statistic	std_error	p_value	conf_low	conf_high
secf_2R211	NA	NA	NA	NA	NA	NA
secf_2R212	-0.2337	-6.9291	0.0337	0.0000	-0.2998	-0.1676
secf_2R213	-0.0843	-2.0787	0.0406	0.0377	-0.1638	-0.0048
secf_2R215	0.2254	3.0426	0.0741	0.0024	0.0802	0.3707
secf_2R216	0.3932	2.2543	0.1744	0.0242	0.0513	0.7350
secf_2R217	NA	NA	NA	NA	NA	NA
secf_2R218	-0.9770	-5.7194	0.1708	0.0000	-1.3118	-0.6421
secf_2R219	0.3955	2.0791	0.1902	0.0376	0.0226	0.7684
secf_2R220	-0.1946	-3.6621	0.0532	0.0003	-0.2988	-0.0905
secf_2R221	-0.2742	-6.9521	0.0394	0.0000	-0.3515	-0.1969
secf_2R222	-0.5294	-11.8776	0.0446	0.0000	-0.6168	-0.4420
secf_2R223	-0.1996	-3.4919	0.0572	0.0005	-0.3117	-0.0876
secf_2R224	-0.6330	-11.1773	0.0566	0.0000	-0.7440	-0.5220
secf_2R225	-0.2525	-5.2611	0.0480	0.0000	-0.3466	-0.1584
secf_2R226	NA	NA	NA	NA	NA	NA
secf_2R228	-0.6602	-1.7527	0.3767	0.0797	-1.3986	0.0782
secf_2R229	-0.7141	-1.8950	0.3769	0.0581	-1.4529	0.0246
secf_2R230	NA	NA	NA	NA	NA	NA
secf_3R301	-0.8016	-3.6798	0.2179	0.0002	-1.2287	-0.3746
secf_3R303	-0.6198	-14.1804	0.0437	0.0000	-0.7055	-0.5341
secf_3R304	-0.7640	-6.3654	0.1200	0.0000	-0.9993	-0.5287
secf_3R305	NA	NA	NA	NA	NA	NA
secf_3R306	0.2733	2.2671	0.1205	0.0234	0.0370	0.5096
secf_3R307	-0.5070	-13.5933	0.0373	0.0000	-0.5801	-0.4339
secf_3R308	-0.9409	-7.8485	0.1199	0.0000	-1.1760	-0.7059
secf_3R309	-0.7612	-4.9307	0.1544	0.0000	-1.0638	-0.4585
secf_3R310	0.2887	4.3852	0.0658	0.0000	0.1596	0.4177
secf_3R311	-0.4250	-3.7086	0.1146	0.0002	-0.6497	-0.2004
secf_3R312	-0.7230	-12.3243	0.0587	0.0000	-0.8381	-0.6080
secf_3R313	-0.6712	-8.9077	0.0754	0.0000	-0.8189	-0.5235
secf_3R314	-0.3732	-5.4910	0.0680	0.0000	-0.5064	-0.2400
secf_3R315	-0.3780	-6.7507	0.0560	0.0000	-0.4877	-0.2682
secf_3R316	-0.7878	-8.5244	0.0924	0.0000	-0.9690	-0.6067
secf_3R317	-0.8866	-3.3278	0.2664	0.0009	-1.4090	-0.3643
secf_3R318	-0.3758	-1.7252	0.2178	0.0845	-0.8028	0.0512
secf_3R319	-0.6491	-20.2565	0.0320	0.0000	-0.7119	-0.5863
secf_3R320	-0.5269	-13.6066	0.0387	0.0000	-0.6028	-0.4510
secf_3R321	-1.1093	-9.7087	0.1143	0.0000	-1.3333	-0.8853
secf_3R322	-0.3529	-7.5706	0.0466	0.0000	-0.4443	-0.2615
secf_3R323	-0.5066	-13.8687	0.0365	0.0000	-0.5783	-0.4350
secf_3R324	-0.8665	-8.5254	0.1016	0.0000	-1.0657	-0.6672
secf_3R325	-0.5899	-11.3201	0.0521	0.0000	-0.6921	-0.4877
secf_3R326	-0.9796	-5.1913	0.1887	0.0000	-1.3495	-0.6097
secf_4R401	-0.7018	-15.2868	0.0459	0.0000	-0.7917	-0.6118
secf_4R402	-1.3689	-14.6178	0.0936	0.0000	-1.5525	-1.1853
secf_4R403	-0.5701	-10.8890	0.0524	0.0000	-0.6727	-0.4674
secf_4R404	-0.4117	-7.6226	0.0540	0.0000	-0.5176	-0.3058
secf_4R405	-0.4485	-13.6011	0.0330	0.0000	-0.5131	-0.3839
secf_4R406	-0.5160	-15.7255	0.0328	0.0000	-0.5804	-0.4517
secf_4R407	-0.6075	-8.9673	0.0677	0.0000	-0.7403	-0.4747
secf_4R408	-0.6058	-15.5868	0.0389	0.0000	-0.6820	-0.5296
secf_4R409	-0.4848	-7.3410	0.0660	0.0000	-0.6143	-0.3554
secf_4R410	-0.2202	-7.2438	0.0304	0.0000	-0.2798	-0.1606
secf_4R411	-0.7070	-16.4548	0.0430	0.0000	-0.7913	-0.6228
secf_4R412	-0.6239	-6.8884	0.0906	0.0000	-0.8014	-0.4463
secf_4R413	-0.3539	-9.1123	0.0388	0.0000	-0.4301	-0.2778
secf_4R416	-0.9122	-2.4217	0.3767	0.0155	-1.6506	-0.1738

term	estimate	statistic	std_error	p_value	conf_low	conf_high
secf_4R417	-0.2273	-6.2777	0.0362	0.0000	-0.2982	-0.1563
secf_4R419	-0.9542	-12.2276	0.0780	0.0000	-1.1072	-0.8013
secf_4R421	-0.9009	-11.0841	0.0813	0.0000	-1.0602	-0.7415
secf_5R501	-0.7956	-11.3905	0.0698	0.0000	-0.9325	-0.6587
secf_5R502	-0.6587	-8.8304	0.0746	0.0000	-0.8050	-0.5125
secf_5R503	0.3902	2.0361	0.1917	0.0418	0.0145	0.7659
secf_5R504	0.2503	2.3694	0.1056	0.0178	0.0432	0.4574
secf_5R505	NA	NA	NA	NA	NA	NA
secf_5R509	0.4061	1.8561	0.2188	0.0635	-0.0228	0.8350
secf_5R510	NA	NA	NA	NA	NA	NA
secf_5R511	NA	NA	NA	NA	NA	NA
secf_5R514	0.3501	2.8943	0.1210	0.0038	0.1130	0.5872
secf_5R517	0.6333	2.3588	0.2685	0.0184	0.1070	1.1597
secf_5R523	-0.7427	-2.7833	0.2668	0.0054	-1.2657	-0.2196
secf_5R524	NA	NA	NA	NA	NA	NA
secf_5R528	-0.6191	-2.3202	0.2668	0.0204	-1.1422	-0.0960
secf_5R530	0.3838	1.7582	0.2183	0.0788	-0.0441	0.8118
secf_5R533	-0.8273	-3.0775	0.2688	0.0021	-1.3543	-0.3003
secf_5R534	-0.3351	-1.7615	0.1903	0.0782	-0.7081	0.0378
secf_5r538	NA	NA	NA	NA	NA	NA
secf_6R601	-1.2688	-7.4984	0.1692	0.0000	-1.6005	-0.9371
secf_6R602	-0.9656	-5.1081	0.1890	0.0000	-1.3361	-0.5950
secf_6R603	-1.2409	-9.2305	0.1344	0.0000	-1.5045	-0.9774
secf_6R604	-0.3909	-4.5140	0.0866	0.0000	-0.5606	-0.2211
secf_6R605	-0.2482	-2.0642	0.1202	0.0390	-0.4839	-0.0125
secf_6R606	-0.5162	-2.3659	0.2182	0.0180	-0.9440	-0.0885
secf_6R609	NA	NA	NA	NA	NA	NA
secf_6R611	-0.2424	-3.3288	0.0728	0.0009	-0.3851	-0.0996
secf_6R612	-0.2625	-1.6961	0.1548	0.0899	-0.5660	0.0409
secf_6R614	NA	NA	NA	NA	NA	NA
secf_6R618	NA	NA	NA	NA	NA	NA
secf_6R620	-0.7340	-14.9096	0.0492	0.0000	-0.8306	-0.6375
secf_7R703	-0.5338	-5.8985	0.0905	0.0000	-0.7112	-0.3564
secf_7R704	-0.3871	-12.0561	0.0321	0.0000	-0.4500	-0.3242
secf_7R705	-0.4051	-17.3841	0.0233	0.0000	-0.4508	-0.3594
secf_7R706	-0.5245	-4.5817	0.1145	0.0000	-0.7488	-0.3001
secf_7R707	-0.6317	-2.9033	0.2176	0.0037	-1.0583	-0.2052
secf_7R708	-0.5602	-10.8560	0.0516	0.0000	-0.6613	-0.4590
secf_7R709	-0.4120	-9.0964	0.0453	0.0000	-0.5008	-0.3232
secf_7R710	-0.2269	-4.9635	0.0457	0.0000	-0.3166	-0.1373
secf_7R711	-0.5949	-13.9942	0.0425	0.0000	-0.6782	-0.5116
secf_7R712	-0.4550	-4.9128	0.0926	0.0000	-0.6365	-0.2734
secf_7R713	-0.5054	-7.3026	0.0692	0.0000	-0.6411	-0.3698
secf_7R714	NA	NA	NA	NA	NA	NA
secf_7R715	NA	NA	NA	NA	NA	NA
secf_7R716	-0.6915	-6.0263	0.1147	0.0000	-0.9164	-0.4665
secf_7R717	-1.1154	-15.4746	0.0721	0.0000	-1.2566	-0.9741
secf_7R718	-0.8827	-14.4522	0.0611	0.0000	-1.0024	-0.7630
secf_7R719	-0.3468	-11.2151	0.0309	0.0000	-0.4074	-0.2862
secf_7R720	-0.6481	-2.4334	0.2663	0.0150	-1.1703	-0.1260
secf_7R722	-0.8069	-3.0271	0.2665	0.0025	-1.3294	-0.2844
secf_7R724	-0.6294	-4.0762	0.1544	0.0000	-0.9320	-0.3267
secf_CGRS6	NA	NA	NA	NA	NA	NA

